Finding and Certifying Weighted Minimum Cuts with Cut Queries

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Abstract

We study the deterministic and nondeterministic cut query complexity of minimum cut in undirected graphs with arbitrary nonnegative edge weights. For the nondeterministic case, we show that every n-vertex graph has an $O(n \log^2(2n))$-query certificate of its exact global minimum-cut value, such that any other nonnegatively weighted graph on the same vertex set that agrees on these queries has the same minimum cut value. A key ingredient is a way to certify a maximum s–t flow value with only $O(n)$ cut queries.

For the deterministic case we give an algorithm that finds an exact global minimum cut using $O(n^{3/2} \log^3(2n)(\log\log(2n))^2)$ cut queries. Previously no $O(n^{2-\varepsilon})$ deterministic bound was known for arbitrary nonnegatively weighted graphs. The algorithm follows Karger's two-respecting tree approach. The key new ingredient is a deterministic construction of a weak skeleton of the input graph G: an explicit weighted graph H in which every minimum cut of G is a 5/4-approximate minimum cut.

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1 Introduction

In this paper we study the problem of finding the global minimum-cut value of an undirected graph with arbitrary nonnegative real edge weights using cut queries. A cut query to a graph $G = (V, w)$ specifies some subset $S \subseteq V$ and returns the total weight $c_G(S) = w(S, V \setminus S)$ of edges with exactly one endpoint in S. Write $n = |V| \geq 2$ and $\lambda(G) = \min_{\varnothing \neq S \subsetneq V} c_G(S)$. We give new upper bounds for computing $\lambda(G)$ in both the deterministic and nondeterministic cut query models. As usual in the cut query setting, we count only oracle calls; exact arithmetic and offline computation are unrestricted.

Minimum cuts have repeatedly shown themselves to have a great deal of structure that allows for efficient descriptions. One result of this form concerns the *cut dimension*. For a graph G with m positive-weight edges, represent each minimum cut by the characteristic vector of its edges in $\{0,1\}^m$. The cut dimension of G is the dimension of the span of these vectors over all minimum cuts. Graur, Pollner, Ramaswamy, and Weinberg [GPRW20] introduced the cut dimension as a lower-bound technique for the deterministic cut-query complexity of global minimum cut. They proved an $n+1$ upper bound for the analogous dimension of minimum s–t cuts, even in directed weighted graphs, but left open the question of how large the cut dimension of global minimum cuts could be. Lee, Li, Santha, and Zhang [LLSZ21] then showed that the cut dimension of global minimum cuts is at most $2n-3$, and that this bound is tight. Faced with this obstruction to proving larger lower bounds, they introduced the ℓ_1-approximate cut dimension, a generalization with the potential to give stronger lower bounds. They showed an infinite family of n-vertex graphs, with $n = 3k+1$, whose ℓ_1-approximate cut dimension is $2n-2$ and left showing larger lower bounds as an open question.

Later, Auza and Lee [AL21] showed that, up to an additive term of at most one, the ℓ_1-approximate cut dimension equals the *linear-query* certificate complexity of minimum cut. A linear query is more general than a cut query and may ask for any linear combination of the edge weights. Here we study certificates restricted to cut queries, resulting in a potentially even larger quantity that is still a lower bound on deterministic cut query complexity. A *cut-query certificate* for a graph G is a set of queries $\mathcal{Q}$ such that every nonnegatively weighted graph G' on the same vertex set that agrees with G on all queries in $\mathcal{Q}$ has the same minimum-cut value. The certificate queries may be chosen with full knowledge of G. Certifying that the minimum-cut value is at most $\lambda(G)$ is easy: just query a shore realizing it. The hard part is certifying that no shore can achieve a smaller value.

We show that the cut-query certificate complexity of minimum cut is $O(n \log^2(2n))$. Consequently, neither cut-query certificate complexity nor ℓ_1-approximate cut dimension can yield a larger deterministic query lower bound for minimum cut. This is yet another way in which minimum cuts are simple: a nearly-linear number of cut queries suffices to fix the minimum cut value in a graph. The key ingredient in our proof is a way to certify the existence of an s–t flow of a given feasible value using just $O(n)$ cut queries.

With another lower bound route blocked off, we turn to deterministic algorithms for finding a global minimum cut. In the randomized case, this question has essentially been solved. We know randomized $\widetilde{O}(n)$ cut query algorithms for both simple unweighted graphs [RSW18] and weighted graphs [MN20]; for simple graphs the bound was improved to $O(n)$ in [AEG+22].

In the deterministic case, a simple graph can be reconstructed with $O(n^2/\log n)$ cut queries by simulating the additive queries of Grebinski and Kucherov [GK00], and this was the best bound known until Anand, Saranurak, and Wang [ASW25] gave an $\widetilde{O}(n^{5/3})$ algorithm. Kenneth-Mordoch [KM26] recently improved this to $O(n^{3/2}\log^2(2n))$ cut queries.

For arbitrary nonnegative real weights, however, the best previously known upper bound was the trivial one from reconstructing the graph, which uses $\Theta(n^2)$ cut queries, far from the lower

bound of $2n-2$. We give a deterministic algorithm using $O(n^{3/2}\log^3(2n)(\log\log(2n))^2)$ cut queries for the weighted case, within a $\log(2n)(\log\log(2n))^2$ factor of the current best unweighted bound. With the same number of queries the algorithm can return every minimum cut of G, not just one.

The recent deterministic algorithms for simple graphs [ASW25, KM26] use maximum-flow and minimum s–t-cut algorithms. Our algorithm instead follows the two-respecting-cut approach of Karger [Kar00]. Karger's nearly-linear time randomized algorithm finds $O(\log n)$ spanning trees of the input graph such that, with high probability, some minimum cut contains at most two edges of one of these trees; such a cut *two-respects* the tree. This reduces the problem to finding a minimum cut among those that two-respect a given tree. Gawrychowski, Mozes, and Weimann [GMW21], building on Mukhopadhyay and Nanongkai [MN20], gave a simple deterministic algorithm for this fixed-tree problem. While they did not explicitly state the algorithm in the cut query model, it is easily adapted to this model: we show in Section 12 how to implement their reduction with $O(n\log n)$ deterministic cut queries, even when the supplied tree has edges absent from the input graph. Thus the remaining problem is to find a small family of trees such that a minimum cut two-respects at least one of them.

Karger obtains the trees by sparsifying the graph with random sampling and using a spanning-tree packing [Kar00]. We do not know how to construct such a sparsifier with a comparable deterministic cut-query bound. Instead, we construct a weaker object, a *weak skeleton*. After handling zero minimum cuts separately, we use $O(n^{3/2}\log^3(2n)(\log\log(2n))^2)$ cut queries to construct an explicit nonnegative weighted graph H on $V(G)$ such that $\lambda(H)\geq\lambda(G)$ and every minimum cut shore S^* of G satisfies $c_H(S^*)\leq\frac{5}{4}\lambda(G)$. In particular, every minimum cut shore S^* of G is a $5/4$-approximate minimum cut of H. We then construct a family of $O(\log n)$ spanning trees of H such that every $5/4$-approximate minimum cut of H crosses at most two edges of some tree in the family (Lemma 17). This step uses only the explicit graph H and requires no further queries to G. Searching these trees with cut queries to G finds an exact minimum cut in another $O(n\log^2 n)$ queries.

Section 2 gives an overview of the certificate proof. Section 3 gives a roadmap of the deterministic algorithm, and Section 4 and Section 5 explain how we construct a weak skeleton via a tree cut sparsifier. Complete proofs are given in Sections 6 to 12.

Lean Formalization As part of this work we have developed a library for cut-query algorithms in Lean 4, available at `https://github.com/troyjlee/cut-query-complexity`. The library formalizes the main certificate and deterministic cut-query theorems of this paper together with their central ingredients, as well as many previous cut-query results on global minimum cut, s–t minimum cut and all-pairs minimum cut. We hope that this library will make it easier for others working on cut-query algorithms to formalize their results in Lean. Appendix A records the correspondence between the named results of this paper and the Lean statements, including differences in formulation and the few claims that are not formalized.

2 Certifying a lower bound with few cut values

Define $\mathrm{cert}(G)$ to be the minimum size of a family $\mathcal{Q}$ of cuts such that

$$c_{G'}(Q)=c_G(Q)\quad(Q\in\mathcal{Q}),\qquad w_{G'}\geq 0\quad\Longrightarrow\quad\lambda(G')=\lambda(G). \tag{1}$$

The competing graph G' has the same vertex set and may put positive weight on any pair, including a nonedge of G. Equivalently, $\mathrm{cert}(G)$ is the nondeterministic cut-query complexity of determining $\lambda(G)$ on input G: the query shores may be provided by an all-knowing prover with knowledge of G, but their answers must determine the minimum-cut value uniquely. In particular $\mathrm{cert}(G)$

is a lower bound on the number of cut queries that any deterministic algorithm computing λ on nonnegatively weighted graphs makes on input G: the shores it asks on G are answered identically by every consistent G', so its output, which is $\lambda(G')$, equals $\lambda(G)$.

Theorem 1 (Near-linear certificates)**.** *Every nonnegatively weighted undirected graph G on $n \geq 2$ vertices satisfies*

$$\mathrm{cert}(G) = O(n \log^2(2n)).$$

The certificate can be chosen to contain a shore S^* that remains a minimum cut in every consistent nonnegatively weighted graph on $V(G)$. If $\lambda(G) = 0$, the single query S^* suffices.

Fix $\lambda = \lambda(G)$ and query an optimum shore for the upper bound. If $\lambda = 0$, this already certifies the optimum by nonnegativity. Otherwise, the difficulty is to force every nontrivial cut to have value at least λ.

For s–t cuts there is a clear strategy to lower bound their value: a feasible s–t flow of value c certifies that every s–t cut has value at least c. A first thought is to exhibit such a flow edge-by-edge, but in some weighted graphs every maximum s–t flow uses $\Omega(n^2)$ edges, making this strategy too expensive.

Our key ingredient is a way to certify the existence of a feasible s–t flow of value c using $O(n)$ cut queries. We describe the flow implicitly through an optimal dual solution to an auxiliary minimum-cost flow problem. The dual solution is represented by one potential per vertex, and a carefully chosen tree T encodes these potentials. Complementary slackness then forces every non-tree edge to carry its full capacity in a direction determined by T. We learn the total incoming and outgoing non-tree flow at each vertex using a constant number of cut queries. These totals determine the remaining divergence that the tree must supply so that the full flow has divergence c at s, $-c$ at t, and zero elsewhere. We explicitly learn the tree-edge capacities with $O(n)$ cut queries and check that the uniquely determined tree flow respects them. We describe this certificate in the next subsection.

We certify that every nontrivial cut has value at least λ by combining the flow certificate with the isolating-cut framework of Li and Panigrahi [LP25]. The obstacle in adapting their terminal reduction is that it relies on expansion guarantees, which we do not know how to certify cheaply with cut queries. We instead use a terminal-halving argument whose key guarantee is witnessed by a feasible flow, so each halving step can be certified with $O(n)$ cut queries. Specifically, given a terminal set U, we retain at most half its terminals in $R \subseteq U$ and certify a feasible flow sending $\lambda/2$ units from each terminal in $U \setminus R$ to R, thereby certifying a lower bound of λ for every shore avoiding R that contains at least two terminals of U. Our construction is described in Section 2.2.

2.1 A tree describes a certifiable flow

A *signed flow* on V is a function $f : V \times V \to \mathbb{R}$ satisfying $f_{uv} = -f_{vu}$ for all $u, v \in V$, where $f_{uv} > 0$ means flow from u to v. The flow is *feasible* in G if $|f_{uv}| \leq w_{uv}$ for all $u, v \in V$. Its *divergence* at a vertex v is the net outflow

$$b_v = \sum_{u \in V} f_{vu},$$

the sum over u of the flow from v to u, so that a source has positive and a sink negative divergence. Every feasible flow satisfies $c_G(S) \geq |\sum_{v \in S} b_v|$ for every cut.

The key novelty in our global mincut certificate is the following lemma.

Lemma 2 (Flow certificates)**.** *If a divergence vector $b \in \mathbb{R}^V$ admits a feasible flow in G, then some labelled spanning tree T of the complete graph on V determines at most $4n - 1$ cut queries certifying*

that b admits a feasible flow in every consistent graph. The queried shores depend only on T and the fixed vertex labels. In particular, an s–t minimum-cut shore and value have a certificate of at most $4n$ queries. For fixed disjoint nonempty terminal sets $A, B \subseteq V$, their minimum separating-cut value has an $O(n)$ cut-query certificate.

To prove Lemma 2, label the vertices $0, \dots, n-1$. For each pair $\{u, v\}$ with $u < v$, assign the auxiliary cost $a_{uv} = 2^{un+v}$, and put $a_{vu} = -a_{uv}$. The sum of the costs around any simple cycle is nonzero for every choice of signs: the largest power of two exceeds the sum of all smaller powers. We then choose a feasible flow f with divergence b minimizing $\sum_{u<v} a_{uv} f_{uv}$.

By complementary slackness for minimum-cost flows (see Bertsekas [Ber91, Sections 1.2.3 and 2.3]), there are vertex potentials p_v such that, for the *reduced costs* $\rho_p(u, v) = a_{uv} - p_u + p_v$, a negative reduced cost forces $f_{uv} = w_{uv}$ and a positive reduced cost forces $f_{uv} = -w_{uv}$. Thus every unsaturated edge has zero reduced cost. The pairs with zero reduced cost form a forest: around any cycle of such pairs, the identities $a_{uv} = p_u - p_v$ would telescope to a zero signed cost sum, contrary to our choice of costs.

The vertex potentials have one additive degree of freedom on each component of the zero-reduced-cost forest: a common shift preserves all potential differences, and hence all reduced costs, within that component. We use this freedom to extend the forest to a spanning tree while preserving complementary slackness. Choose a component and a pair crossing it with minimum absolute reduced cost, and add the same constant to every vertex potential in the component until that pair has reduced cost zero. By this choice of crossing pair, no reduced cost reverses sign, so f still satisfies the same optimality conditions. Each step merges components; repeating produces a spanning tree T of pairs with zero reduced cost. Root T at any vertex r and normalize $p_r = 0$. The equations $p_u - p_v = a_{uv}$ along tree edges then determine all vertex potentials, and hence every non-tree reduced cost. Thus T encodes the direction in which each non-tree pair is saturated. This applies even to pairs absent from G, and T may itself contain zero-capacity edges. The full construction and proof appear in Section 7.1.

It remains to check that this prescribed non-tree flow can be completed on T to have divergence b. First query all singleton cuts and the two-vertex cuts for tree edges. By the well-known fact that for two disjoint sets A, B

$$2w_G(A, B) = c_G(A) + c_G(B) - c_G(A \cup B), \tag{2}$$

these $n + (n-1)$ queries determine the capacity of every tree edge. For each vertex v, let O_v be its outgoing non-tree neighbors as prescribed by T, and query O_v and $O_v \cup \{v\}$. The same identity gives the total outgoing non-tree flow at v. Its total incoming non-tree flow is the singleton cut value minus the incident tree capacities and the outgoing non-tree flow. Thus at most $n + (n-1) + 2n = 4n - 1$ queries determine all tree capacities and the non-tree divergence b^{nt}.

The residual divergence $q = b - b^{\mathrm{nt}}$ sums to zero and determines a unique flow on T. If deleting a tree edge uv leaves the component A containing u, its flow from u to v must be $\sum_{x \in A} q_x$. Checking the absolute values of these $n-1$ flows against their measured tree capacities completes verification. Figure 1 illustrates this recovery from the tree.

In any graph G' consistent with these answers, the tree capacities and the measured non-tree divergence are unchanged. Saturate its non-tree pairs in the directions fixed by T, using their capacities in G', and add the same tree flow. The resulting flow is feasible and has divergence b. This also handles new positive edges, since T already prescribes a direction for every non-tree pair.

For the s–t conclusion of Lemma 2, certify a maximum flow and query a matching minimum-cut shore. This gives both bounds on the optimum using at most $4n$ queries. If the optimum is zero, the shore query alone suffices by nonnegativity.

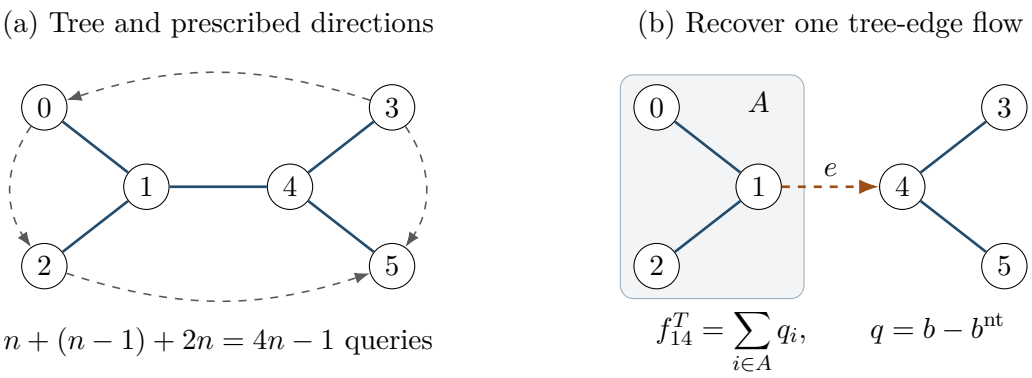


Figure 1: A flow certificate described by a tree. Solid edges form T; dashed arrows in (a) show selected saturated non-tree pairs. Deleting $e = \{1, 4\}$ leaves the component $A = \{0, 1, 2\}$. Its residual divergence determines the signed tree flow from 1 to 4; b^{nt} denotes the non-tree divergence.

2.2 Isolating cuts and terminal halving

A cut *separates* a terminal set U if each shore contains at least one vertex of U. Start with $U = V$, so every nontrivial cut separates U. We maintain that any nontrivial cut of value below λ in a consistent graph must separate U. Each round has two steps. First, we certify that every cut isolating a single terminal of U has value at least λ. Any remaining cut below λ therefore has at least two terminals on each shore. Second, we certify a feasible flow that forces every cut of value below λ with at least two terminals of U on each shore to separate a smaller terminal set $R \subseteq U$, with $|R| \le |U|/2$. We then replace U by R and repeat. Eventually U is a singleton, which no cut can separate, establishing the desired lower bound.

Certifying isolating cuts. For a terminal set U and $v \in U$, an isolating cut separates v from $U \setminus \{v\}$. We certify all these optimum values using $O(n \log |U|)$ queries (Lemma 11). We combine the isolating-cut framework of Li and Panigrahi [LP25] with the flow certificates above. Give the terminals distinct binary labels. For each bit, query a minimum cut separating its zero and one classes and certify its optimality by a flow. Intersect the chosen shores containing a terminal v to obtain a cell P_v. The cells are disjoint, and an uncrossing argument places an inclusion-minimal optimum isolating shore inside P_v. Because the bit cuts are certified, this containment holds in every consistent graph.

We can now certify the local flow problem in every cell. Contract the outside of P_v to a single sink t_v. A cut query to the contracted graph is one cut query to G: a shore $X \subseteq P_v$ has value $c_G(X)$, and a shore containing t_v has value $c_G(P_v \setminus (X \setminus \{t_v\}))$. A consistent graph contracts to a consistent local graph. The total size of these local networks is $O(n)$ because the cells are disjoint, so their flow certificates and optimum-shore queries cost $O(n)$ altogether; the $\lceil \log_2 |U| \rceil$ bit cuts account for the logarithm. This is where the isolating-cut framework makes many lower-bound witnesses affordable.

Halving the terminal set. For $|U| \ge 2$, there is a nonempty $R \subset U$, $|R| \le |U|/2$, such that every discarded terminal simultaneously sends $\lambda/2$ units to R, with zero divergence outside U (Lemma 13). Certifying this divergence forces

$$c_{G'}(S) \ge \frac{\lambda}{2} |S \cap U| \qquad \text{whenever } S \cap R = \varnothing. \tag{3}$$

To obtain the flow, we split off the nonterminals. For multigraphs, the splitting-off theorems of Lovász and Mader [Lov76, Mad78] allow this: a nonterminal can be removed, replacing pairs of its edges su, sv by edges uv, without lowering the edge connectivity between any two remaining vertices. We need a fractional form for arbitrary real capacities, which we prove directly (Lemma 12). The resulting graph J is supported on U, is dominated by G on every cut, and retains value at least λ on every U-separating cut. Choose $X \subseteq U$ maximizing $c_J(X)$ and let R be the smaller of X and $U \setminus X$. Every terminal outside R has at least half its incident weight going to R, since otherwise moving it across would improve the maximum cut. Its weighted degree is at least λ. It follows that $c_G(S) \geq c_J(S) \geq (\lambda/2)|S \cap U|$ for every shore avoiding R. Max-flow/min-cut turns these inequalities into the simultaneous flow in G. Only its divergence is certified; the split-off graph and maximum cut are chosen with full input knowledge.

Equation (3) shows that a cut below λ with at least two terminals of U on each shore must separate R: a shore avoiding R would have value at least λ.

Each of the at most $\lceil \log_2 n \rceil$ rounds costs $O(n \log |U|)$ queries for isolating cuts and $O(n)$ for the halving flow, giving $O(n \log^2(2n))$ in total. If $\lambda = 0$, the queried optimum shore alone suffices by nonnegativity.

3 The deterministic algorithm: a roadmap

The cut certificate for G of Section 2 already yields a *skeleton*, a sparse graph H with $O(n \log^2 n)$ edges and the same vertex set as G such that $\lambda(H) = \lambda(G)$ and a shore S^* that realizes a minimum cut of G also realizes a minimum cut of H (Proposition 15). The certificate, however, is chosen with full knowledge of G. For our deterministic algorithm we settle for a weaker object.

Definition 3 (Weak skeleton)**.** Let G be a graph with $\lambda = \lambda(G) > 0$. An explicit nonnegative weighted graph H on $V(G)$ is a *weak skeleton* of G if $\lambda(H) \geq \lambda$ and every minimum-cut shore S^* of G satisfies $c_H(S^*) \leq \frac{5}{4}\lambda$. In particular every minimum cut of G is a 5/4-approximate minimum cut of H.

Once H is explicit, we follow Karger's nearly linear-time algorithm [Kar00]: a tree-packing argument deterministically produces $O(\log n)$ spanning trees of H such that every 5/4-approximate minimum cut of H, and in particular S^*, two-respects one of them (Lemma 17). We then adapt the algorithm of Gawrychowski, Mozes, and Weimann [GMW21] to the cut-query model to find a minimum two-respecting cut of G for each tree with $O(n \log n)$ cut queries (Theorem 19 and Section 12). These two steps require only modest adaptations of prior work. The main novelty is deterministically constructing the weak skeleton, which we focus on in this overview.

The pipeline. After testing connectivity of the positive-weight support of G with $O(n \log n)$ queries, and returning a zero cut if it is disconnected, we may assume $\lambda(G) > 0$. The weak skeleton is then constructed in two stages, with a parameter α to be chosen.

(i) *Learn* (Section 5). Using $O((n^2/\alpha) \log^3(2n)(\log\log(2n))^2)$ cut queries, contract vertex pairs that no minimum cut of G separates, and find an explicit tree with $O(n)$ vertices that approximates every cut of the contracted graph within a factor α. Such a tree is an α-tree cut sparsifier; the precise definitions follow in Section 3.1. This stage works for every α between $C \log^2(2n)(\log\log(2n))^2$ and $\sqrt{n}$, for an absolute constant C.

(ii) *Convert* (Section 4). Using $O(n\alpha \log^3(2n)(\log\log(2n))^2)$ further cut queries, turn that tree into a weak skeleton of the contracted graph, and lift it back to G.

Balancing the two stages at $\alpha = \Theta(\sqrt{n})$ gives the following theorem. For the finitely many n with $\sqrt{n}$ below the threshold $C \log^2(2n)(\log\log(2n))^2$, the graph is simply reconstructed with $O(n^2) = O(1)$ queries.

Theorem 4 (Deterministic weighted minimum cut)**.** *There is a deterministic algorithm that, given cut-query access to an n-vertex undirected graph G, $n \geq 2$, with arbitrary nonnegative real edge weights, returns $\lambda(G)$ and every minimum-cut shore of G using*

$$O\Big(n^{3/2} \log^3(2n)(\log\log(2n))^2\Big)$$

cut queries. If $\lambda(G) = 0$, it returns the partition of V into the components of the positive support, whose proper unions are exactly the minimum-cut shores.

Two features of the construction let the algorithm return every minimum cut rather than a single one: a weak skeleton approximates every minimum cut of G at once, and the two-respecting search can report every cut at a known level rather than a single lightest one (Theorem 52). The proof is given in Section 11.8.

The rest of this section fixes the definitions and notation needed for the weak skeleton construction.

3.1 Quotients, hierarchies, and tree cut sparsifiers

Quotients and safe partitions. For a partition $\mathcal{P}$ of V, the *quotient graph* $G/\mathcal{P}$ has one vertex for each part of $\mathcal{P}$. The weight between distinct parts A, B is the total weight $w_G(A, B)$ of edges joining them in G; edges within a part are discarded. Every shore $\mathcal{S} \subseteq \mathcal{P}$ therefore satisfies

$$c_{G/\mathcal{P}}(\mathcal{S}) = c_G\left(\bigcup_{A \in \mathcal{S}} A\right),$$

so one cut query to the quotient is one cut query to G. A partition $\mathcal{P}$ of V is *safe* if no minimum cut of G separates two vertices in the same part. Equivalently, every minimum-cut shore of G is a union of parts of $\mathcal{P}$, and $\lambda(G/\mathcal{P}) = \lambda(G)$. The learning stage contracts only vertex pairs that no minimum cut separates, so its contracted classes form a safe partition.

Tree cut sparsifiers. Let T be a nonnegatively weighted tree with $V(G) \subseteq V(T)$. The vertices in $V(G)$ are the *terminals*; the other vertices of T are *Steiner vertices*. A shore $S \subseteq V(G)$ is represented in T by any vertex set X with $X \cap V(G) = S$, and its *representation cost* is the cheapest such representation,

$$R_T(S) = \min_{\substack{X \subseteq V(T) \\ X \cap V(G) = S}} c_T(X).$$

We drop the subscript when the tree is clear from context. A tree whose representation costs approximate every cut of G is a tree cut sparsifier [ADK26, Definition 2.4].

Definition 5 (Tree cut sparsifier)**.** Let $G = (V, w)$ and $\alpha \geq 1$. A nonnegatively weighted tree T with $V \subseteq V(T)$ is an *α-tree cut sparsifier* of G if

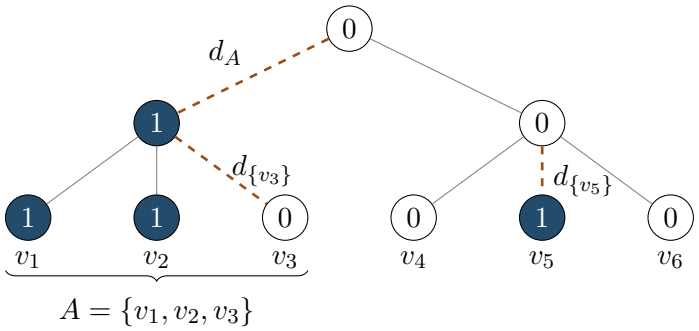
$$c_G(S) \leq R_T(S) \leq \alpha c_G(S) \qquad \text{for every } S \subseteq V. \tag{4}$$

Hierarchies and canonical capacities. The tree cut sparsifiers we work with have a special form introduced by Räcke [Räc02]. A *hierarchy* of G is a rooted tree T with leaf set $V(G)$ and no unary internal nodes, so that it has fewer than $2n$ nodes. Each nonroot node represents its descendant leaf cluster A. The hierarchy is *canonical* if the parent edge of each such node carries the capacity $d_A = c_G(A)$. Any tree cut sparsifier of G can be made canonical with $O(n)$ queries without weakening either bound in (4) as described in Section 4.

In a hierarchy, representing S means labeling the leaves by $\mathbf{1}_S$ and extending the binary labels to internal nodes; $R_T(S)$ is the minimum total capacity of edges whose endpoint labels disagree. Figure 2 shows the cost of one labeling. Every original edge crossing S crosses at least one of the cluster boundaries in any labeling, so canonical capacities give

$$c_G(S) \le R_T(S) \qquad \text{for every } S \subseteq V(G) \tag{5}$$

with no further assumption on the tree. The algorithm will repeatedly build explicit graphs J from learned data, for instance a learned subgraph of G, and need a canonical hierarchy of J whose representation costs approximate the cuts of J. Since J is explicit, this is an offline computation: it costs no queries, and we place no limit on its running time. In this setting an algorithm of Räcke and Shah [RS14] gives, for every nonnegative weighted graph J on r vertices, a canonical hierarchy with $R(S) \le a(r)\, c_J(S)$ for all S, where $a(r) = O(\log(2r) \log\log(2r))$ (Lemma 28); since offline computation is unrestricted, their construction may use exact rather than approximate sparsest cuts, which is where these logarithms come from.



Cost of this labeling:

$$d_A + d_{\{v_3\}} + d_{\{v_5\}}$$

Minimize over internal labels to obtain $R(S)$.

Figure 2: Representing $S = \{v_1, v_2, v_5\}$ in a hierarchy. Leaf labels are fixed by S; internal labels are chosen freely. Each edge leading to a cluster B has capacity $d_B = c_G(B)$. The dashed edges join unequal labels and contribute to the cost. The displayed labeling is one candidate in the minimization defining $R(S)$.

4 From a tree cut sparsifier to a weak skeleton

The conversion stage proves the following theorem.

Theorem 6 (From a tree cut sparsifier to a weak skeleton)**.** *Given cut-query access to a graph G on n vertices with $\lambda(G) > 0$ and an explicit B-tree cut sparsifier of G with $O(n)$ vertices, there is a deterministic algorithm that constructs a weak skeleton of G using*

$$O\Big(nB \log^3(2n)(\log\log(2n))^2\Big)$$

cut queries.

The full proof is in Sections 9 and 10; this section explains the ideas. Throughout, G denotes the current quotient, with $\lambda = \lambda(G) > 0$, and we are given a B-tree cut sparsifier of G with $O(n)$ vertices. We first bring it into normal form: make every terminal a leaf, contract non-terminal leaves into their neighbours, root the tree at an internal node, and suppress unary nodes, so that it is a hierarchy; then query the cut value $c_G(A)$ of each of its $O(n)$ clusters and make it the canonical capacity d_A. None of these steps increases a representation cost: a labeling of the old tree restricts or extends to a labeling of the new tree of no greater cost, and the edge above a cluster A represents A by itself, so its old capacity is at least $R(A) \geq c_G(A)$. Let T be the resulting canonical hierarchy and write $R(S) = R_T(S)$.

Of the sparsifier guarantee (4), we keep the lower bound $c_G(S) \leq R(S)$ for every shore, which (5) gives for free, and from the upper bound we retain only what it says about minimum cuts:

$$R(S^*) \leq B\lambda \qquad \text{for every minimum-cut shore } S^* \text{ of } G. \tag{6}$$

We call (6) the *promise* with bound B. Dropping the upper bound for all other cuts is deliberate: the refinement in Section 4.2 produces intermediate trees that satisfy a promise with a smaller B but need not have all-cut distortion bounded by that smaller B. Neither the minimum cuts nor λ are known; all parameters below depend only on the public bound B. The analysis fixes one minimum-cut shore S^*; since the construction never reads S^*, every conclusion drawn for it holds for each minimum cut simultaneously.

We first give a direct construction costing $O(nB^2 \log n)$ queries. We then show that a coarser version of the same construction halves the promise bound with $O(nB \log^3(2n)(\log\log(2n))^2)$ queries, linear in the current bound B. Repeating it while the bound is at least a polylogarithmic threshold, and finishing with the direct construction below that threshold, costs a geometric series dominated by its first term, which gives Theorem 6.

4.1 A direct construction

We want an explicit graph H with $\lambda(H) \geq \lambda$ and $c_H(S^*) \leq \frac{5}{4}\lambda$. We take H to be a weighted sum of two graphs, F and J_{prot}, described next. The role of F is to approximate G: it is learned with cut queries, and its error on a cut S grows with the representation cost $R(S)$, so it is accurate at the minimum cuts, which satisfy the promise, but may undervalue cuts of large representation cost. The role of J_{prot} is to compensate: it is built from the tree without queries, and its cut values track $R(S)$, so it restores every cut that F may undervalue while adding little at S^*.

Ingredient 1: a completion accurate on cheap cuts. Fix an accuracy parameter $0 < \eta < 1$. We learn a nonnegative graph F that agrees with G on every cluster cut and nearly agrees on every symmetric difference of two clusters:

$$c_F(A) = c_G(A), \qquad |c_F(A \triangle B) - c_G(A \triangle B)| \leq 2\eta \min(d_A, d_B) \quad \text{for all clusters } A, B. \tag{7}$$

Clusters are laminar, so $A \triangle B$ is the union of two disjoint clusters or the difference of two nested ones. Writing L_G for the Laplacian, so that $c_G(S) = \mathbf{1}_S^{\mathsf{T}} L_G \mathbf{1}_S$, and expanding $c_G(A \cup B)$ or $c_G(B \setminus A)$ as a quadratic form gives

$$\mathbf{1}_A^{\mathsf{T}} L_G \mathbf{1}_B = \begin{cases} \frac{1}{2}(c_G(A \cup B) - c_G(A) - c_G(B)) & A \cap B = \varnothing, \\ \frac{1}{2}(c_G(A) + c_G(B) - c_G(B \setminus A)) & A \subseteq B. \end{cases}$$

Since F matches every cluster cut, (7) says that these *cluster interactions* satisfy $|\mathbf{1}_A^{\mathsf{T}}(L_F - L_G)\mathbf{1}_B| \leq \eta \min(d_A, d_B)$. For each cluster A, the cut value $d_A = c_G(A)$ is the total weight of the edges leaving

A. Charging each such edge to its endpoint inside A, or instead to its endpoint outside A, gives two additive measures on vertices of total mass d_A,

$$\mu_A^{\text{in}}(X) = w_G(A \cap X, V \setminus A), \qquad \mu_A^{\text{out}}(X) = w_G(A, X \setminus A),$$

both queryable using the rectangle identity (2). For fixed A the interaction with B is one of $-\mu_A^{\text{out}}(B)$, $\mu_A^{\text{in}}(B)$, or $d_A - \mu_A^{\text{out}}(B)$, according as B is disjoint from, inside, or contains A. It therefore suffices to learn each boundary measure of A to within ηd_A on every cluster B.

In depth-first leaf order every cluster is an interval of leaves, so each such mass is a difference of two prefix masses, and it suffices to control the prefix function of each boundary measure. Fix a cluster A and one of its two boundary measures, and put $q = \lceil 2/\eta \rceil$. Binary searches locate the q prefixes at which the measure crosses each multiple of d_A/q; recording the prefixes on both sides of each crossing handles a single heavy edge. Any nonnegative measure agreeing with these checkpoint values has every prefix mass within d_A/q, hence every interval mass within $2d_A/q \le \eta d_A$. Each prefix mass costs two cut queries, so the two measures of a cluster cost $O(\eta^{-1} \log n)$ queries and all $O(n)$ clusters cost $O(n\eta^{-1} \log n)$, regardless of the hierarchy's depth. Finally, F is any nonnegative weighted graph whose cluster cuts equal d_A and whose boundary measures take the recorded checkpoint values. These are linear equalities in the weights of F, and G itself satisfies them, so the system is feasible.

The error on a cut is controlled by its representation cost. Fix the minimum labeling of S and let $A_1, \dots, A_m$ be the clusters whose parent edge it cuts, so that $R(S) = \sum_i d_{A_i}$ and, since each $d_{A_i} \ge \lambda$, $m \le R(S)/\lambda$. Walking from the root to a leaf v, the label changes exactly at the cut edges above the clusters containing v, so up to a constant vector $\mathbf{1}_S = \sum_i \sigma_i \mathbf{1}_{A_i}$ with $\sigma_i \in \{-1, 1\}$. Laplacians annihilate constant vectors, so expanding $c(S) = \mathbf{1}_S^{\mathsf{T}} L \mathbf{1}_S$ for F and G and applying (7) to each of the m^2 terms gives

$$|c_F(S) - c_G(S)| = \Big| \sum_{i,j} \sigma_i \sigma_j \, \mathbf{1}_{A_i}^{\mathsf{T}} (L_F - L_G) \mathbf{1}_{A_j} \Big| \le \sum_{i,j} \eta \, d_{A_i} = \eta \, m \, R(S) \le \eta R(S)^2/\lambda. \tag{8}$$

At a minimum cut the promise makes the error at most $\eta B^2 \lambda$. The same bound identifies the difficulty: a cut with a costly representation might become artificially cheap in F. The next ingredient takes care of this.

Ingredient 2: a protection graph to lower bound costly cuts. The second ingredient makes sure that no cut of H dips below λ, the minimum cut value of G. By (8), the cuts at risk are those with large representation cost $R(S)$, so we want to add a graph whose cut values track $R(S)$. Now $R(S)$ is itself a cut function, the minimum cut of the tree T separating the leaves in S from the rest, but T cannot be added to F directly because its internal nodes are not vertices of G. We therefore construct, without queries, an explicit graph J_{prot} on the leaves of T that mirrors the tree's cuts:

$$R(S) \le c_{J_{\text{prot}}}(S) \le \rho R(S) \qquad \text{for every } S, \tag{9}$$

for an absolute constant ρ. By Theorem 3.7 of Englert et al. [EGK+14], the capacitated tree T admits a constant-quality vertex cut sparsifier on its leaves. Since T is explicit, we can choose such a sparsifier deterministically by finite linear feasibility, without further queries to G. Lemma 26 gives a self-contained proof with $\rho = 42$.

Ingredient 3: the combination. Set $H = F + \sqrt{\eta}\, J_{\text{prot}}$. For every nontrivial cut S, write $x = \sqrt{\eta}\, R(S)/\lambda$. Nonnegativity of F together with (8) gives $c_F(S) \ge \max\{0, 1 - x^2\}\lambda$, and (9) gives

$\sqrt{\eta}\, c_{J_{\text{prot}}}(S) \geq x\lambda$. Hence

$$\frac{c_H(S)}{\lambda} \geq \max\{0, 1 - x^2\} + x \geq 1. \tag{10}$$

If $x \geq 1$ the protection term alone pays; if $x < 1$ then $1 - x^2 + x \geq 1$. So $\lambda(H) \geq \lambda$. For the promised optimum, (8) and (9) give

$$\frac{c_H(S^*)}{\lambda} \leq 1 + \eta B^2 + \rho\sqrt{\eta}\, B.$$

Choosing $\eta = 1/(100\rho^2 B^2)$ makes the right side at most $1 + 1/(100\rho^2) + 1/10 \leq 5/4$. As S^* was an arbitrary minimum cut, together with $\lambda(H) \geq \lambda$ this makes H a weak skeleton of G (Definition 3), using $O(n\eta^{-1} \log n) = O(nB^2 \log n)$ queries.

4.2 Refining the promise

With $B = \sqrt{n}$ the direct construction would cost $\Omega(n^2 \log n)$ queries. It spends $\Theta(B^2)$ checkpoints per cluster because it must approximate the optimum to additive $O(\lambda)$: the completion error at S^* is $\eta B^2 \lambda$, which forces $\eta \approx 1/B^2$.

A *refinement* round takes a canonical hierarchy of G that satisfies the promise with bound B and outputs a new canonical hierarchy of G that satisfies the promise with bound $B/2$: every minimum cut, still unidentified, has representation cost at most $B\lambda/2$ in the new tree. Since the output is a hierarchy and not a skeleton, a round asks less of the completion. The completion need only track G at S^* to within a small fraction of $B\lambda$ rather than within λ, so we allow a larger error, $\eta \approx 1/B$, compute an offline hierarchy of the resulting explicit graph, and query the original graph on the new cluster boundaries. The transfer inequality below shows that this halves the promise bound at a cost linear in B. Throughout, $a = a(n) = O(\log(2n) \log\log(2n))$ is the approximation factor of the offline canonical hierarchy of Räcke and Shah: every explicit nonnegative weighted graph on n vertices has a canonical hierarchy with $R(S) \leq a\, c(S)$ for every shore S (Section 3.1).

Claim (Halving step; informal version of Lemma 30)**.** *Given a canonical hierarchy of G with $O(n)$ clusters satisfying the promise (6) with bound $B \geq 1/\gamma$, one can compute a canonical hierarchy of G with $O(n)$ clusters satisfying the promise with bound $B/2$, using $O(na^2 B \log n)$ cut queries, where $\gamma = 1/(100a^2\rho^2)$.*

To see this, build F and J_{prot} as above with $\eta = \gamma/B$, but form $H = F + \gamma J_{\text{prot}}$, with a heavier contribution from J_{prot} than in the final skeleton. Compute an offline hierarchy of this explicit H with factor a; with the H-capacities $c_H(A)$ its representation cost R_H satisfies $R_H(S^*) \leq a\, c_H(S^*)$. Then query the original capacities $c_G(A)$ of its clusters to make it canonical for G. Since $c_H(S) \geq \gamma R(S)$ for every S, the completion error (8) can be written in terms of H alone:

$$c_G(S) \leq c_H(S) + \frac{\eta}{\gamma^2 \lambda}\, c_H(S)^2.$$

Fix a labeling of S^* that is optimal for the H-capacities, let $A_1, \ldots, A_m$ be the clusters it cuts, and put $h_i = c_H(A_i)$, so that $\sum_i h_i = R_H(S^*)$. Reusing the same labeling with the G-capacities, applying the displayed inequality to each A_i, and using $\sum_i h_i^2 \leq (\sum_i h_i)^2$ transfers it to the new representation cost:

$$\frac{R_{\text{new}}(S^*)}{\lambda} \leq \frac{1}{\lambda} \sum_i c_G(A_i) \leq x + \frac{\eta}{\gamma^2} x^2, \qquad x = \frac{R_H(S^*)}{\lambda} \leq a(1 + \eta B^2 + \gamma\rho B). \tag{11}$$

There is again no depth factor. With $\gamma = 1/(100a^2\rho^2)$, $\eta = \gamma/B$, and $B \geq 1/\gamma$, the right side of (11) is at most $B/2$. The cost is that of the completion, $O(n\eta^{-1} \log n) = O(na^2\rho^2 B \log n)$. Intermediate graphs H serve only to produce better hierarchies; they need not be weak skeletons themselves.

Adding up the queries. Repeat the halving step while the promise bound is at least $1/\gamma = O(\log^2(2n)(\log\log(2n))^2)$. The successive costs are geometric in B and sum to $O(na^2B\log n)$. Once the remaining bound is $b < 1/\gamma$, the direct construction costs $O(nb^2\log n) \le O(nB\log n/\gamma) = O(na^2B\log n)$ as well, since $b \le B$. Altogether, converting a supplied hierarchy with bound B costs $O(nB\log^3(2n)(\log\log(2n))^2)$ queries. Every parameter above is computed from the public bound B and known quantities; the unknown S^* and λ enter only the analysis, so the final H is a weak skeleton of G. This completes the overview of Theorem 6.

5 Learning a tree cut sparsifier

The learning stage proves the following theorem.

Theorem 7 (Learning a tree cut sparsifier)**.** *There is an absolute constant C such that, for every nonnegatively weighted graph G on $n \ge 2$ vertices with $\lambda(G) > 0$ and every $C\log^2(2n)(\log\log(2n))^2 \le \alpha \le \sqrt{n}$, a deterministic algorithm uses*

$$O\left(\frac{n^2}{\alpha}\log^3(2n)(\log\log(2n))^2\right)$$

cut queries to find a safe partition $\mathcal{P}$ of $V(G)$ and an explicit α-tree cut sparsifier of the quotient $G/\mathcal{P}$ with at most $8n$ vertices.

The full proof is in Section 11.7; this section explains the ideas. The algorithm proceeds in phases, each of which either learns a tree cut sparsifier of the current quotient or safely contracts it to at most half as many vertices. We call this the *sparsify-or-shrink* principle. After describing the weighted degree normalization applied in every phase, we state the dichotomy as a lemma conditional on an auxiliary graph (Section 5.2), then show how to learn that auxiliary graph with cut queries (Section 5.3), and finally add up the queries.

Throughout, G denotes the current quotient, N its number of vertices, and $\lambda = \lambda(G) > 0$. We maintain a known nontrivial cut of value $\Lambda \ge \lambda$.

5.1 Weighted degree normalization

Let d_v be the *weighted degree* of vertex v, i.e., the sum of the weights of the edges incident to v. All weighted degrees are learned once, with n cut queries at the start of the algorithm, and are then maintained without further queries, since every contraction we perform is of an edge whose weight we know. We also maintain $\Lambda \le \min_v d_v$: initially Λ is the minimum weighted degree, witnessed by a singleton cut, and Λ only ever decreases, to the weighted degree of a newly contracted vertex or to the value of a newly found nontrivial cut. At the start of each phase, and after every change to Λ, we normalize the current quotient so that its weighted degrees satisfy

$$\lambda \le \Lambda \le d_v \le (N-1)\Lambda \quad \text{for every } v. \tag{12}$$

Whenever a vertex has $d_v > (N-1)\Lambda$, some edge incident to v has weight greater than $\Lambda \ge \lambda$. No minimum cut can cross such an edge, so contracting it is safe. We find such an edge with $O(\log N)$ queries by binary search, repeatedly retaining the half of the candidate neighbors with larger average incident edge weight. If the heavy edge is $\{u, v\}$, contracting u, v gives new weighted degree $d_u + d_v - 2w_G(u, v)$; all other weighted degrees stay unchanged. Decrease N by one, lower Λ to the new vertex's weighted degree if that is smaller, and repeat until (12) holds.

The invariant (12) is used in three ways. It makes the degree spread polynomial in N, so that later scan counts and bucket counts are $O(\log N)$ regardless of the smallest positive edge weight. It forces the vertex count to halve whenever Λ drops by a factor $4N$ (Section 5.2). And it is cheap: weighted degrees are queried only once, every subsequent search costs $O(\log n)$ queries and contracts a pair, and there are at most $n-2$ contractions, so all normalization calls together cost $O(n \log n)$ queries. Every original minimum cut survives.

5.2 Sparsify or shrink

A phase with parameter $2 \le k < N$ seeks a tree cut sparsifier of quality

$$B = O(k \log^2(2N)(\log\log(2N))^2).$$

Suppose we have an explicit auxiliary graph J_{aux} on at most $2N$ vertices whose terminals are the vertices of G and which may have additional vertices. For a shore $S \subseteq V(G)$, define its minimum extension cost in J_{aux} by

$$\Psi(S) = \min_{\substack{X \subseteq V(J_{\text{aux}}) \\ X \cap V(G) = S}} c_{J_{\text{aux}}}(X).$$

The approximation guarantee we require has both a multiplicative and an additive error:

$$c_G(S) \le \Psi(S) \le A c_G(S) + \varepsilon, \qquad A = O(k \log(2N) \log\log(2N)), \qquad \varepsilon = \frac{\Lambda}{8N}. \tag{13}$$

Claim (Dichotomy; informal version of Lemma 40)**.** *Given J_{aux} satisfying (13), one phase either outputs a B-tree cut sparsifier of G with $B = O(k \log^2(2N)(\log\log(2N))^2)$ using $O(N)$ further queries, or safely contracts G to at most $\lfloor N/2 \rfloor$ vertices.*

Compute offline the minimum cut of J_{aux} that separates two vertices of G:

$$\mu = \min_{\varnothing \ne S \subsetneq V(G)} \Psi(S), \qquad \Psi(S_{\text{aux}}) = \mu.$$

Requiring S to be nontrivial excludes cuts separating only added vertices. Since J_{aux} is explicit, computing μ and S_{aux} requires no queries to G.

Case $\mu \ge 2\varepsilon$: sparsify. Every nontrivial shore S satisfies $2\varepsilon \le \Psi(S) \le A c_G(S) + \varepsilon$. Thus $\varepsilon \le A c_G(S)$, and the additive error is absorbed: $\Psi(S) \le 2A c_G(S)$. Apply the offline construction of Räcke and Shah to the explicit graph J_{aux}, obtaining a hierarchy of all its vertices with distortion $a = O(\log(2N) \log\log(2N))$. Leave the labels of the added vertices free and compress the tree to the terminal set $V(G)$ by contracting non-terminal leaves into their neighbours and, on each terminal-free path of degree-two vertices, all but a minimum-capacity edge; this preserves every minimum extension of the terminal labels. Minimizing the hierarchy inequalities over the free labels then gives a tree T' with $O(N)$ clusters and $\Psi(S) \le R_{T'}(S) \le a\Psi(S)$ for every $S \subseteq V(G)$. Finally use $O(N)$ cut queries to replace each cluster capacity by its G-cut value. This lowers capacities while keeping every G-cut dominated, so the resulting tree, with representation cost R_G under these canonical G-capacities, satisfies

$$c_G(S) \le R_G(S) \le B c_G(S) \quad (S \subseteq V(G)), \qquad B = 2aA = O(k \log^2(2N)(\log\log(2N))^2). \tag{14}$$

This is the desired tree cut sparsifier.

Case $\mu < 2\varepsilon$: shrink. Query S_{aux} to obtain a smaller cut value $\Lambda' = c_G(S_{\text{aux}}) \le \mu < \Lambda/(4N)$, and normalize the vertex degrees again to satisfy (12) with this bound. Suppose an old vertex

remained a singleton in the resulting quotient on $m \le N$ vertices. Its weighted degree would be unchanged and at least the old Λ, whereas normalization gives $d_v \le (m-1)\Lambda' < \Lambda$, even if the bound decreased further. Every final supervertex therefore contains at least two old vertices, so $m \le \lfloor N/2 \rfloor$. Each failed phase halves the vertex count, independently of the weight spread.

Thus within $O(\log n)$ phases we either obtain the desired tree cut sparsifier or reach $N \le k$, when we reconstruct the quotient with $O(N^2)$ queries and compute its tree cut sparsifier offline.

5.3 Learning the auxiliary graph

It remains to construct J_{aux} satisfying (13) with few queries.

Claim (Auxiliary graph; informal version of Lemma 39)**.** *An explicit graph J_{aux} satisfying (13) can be learned with $O(Nk \log^2(2N) + (N^2/k)\log(2N))$ cut queries.*

The weighted rounding obstacle. Learning G exactly costs $\Theta(N^2)$ queries, but the quotient of G on a partition of $V(G)$ into $p = O((N/k)\log N)$ parts costs only $O(p^2)$, and it gives the exact value of every cut whose shore is a union of parts. To estimate $c_G(S)$ for an arbitrary shore S from the quotient we *round* S: replace it by a union $\widehat{S}$ of parts. The two shores disagree on the vertices of $S \triangle \widehat{S}$: those of $S \setminus \widehat{S}$ leave the shore and those of $\widehat{S} \setminus S$ join it. An edge changes between cut and uncut only if exactly one of its endpoints is in $S \triangle \widehat{S}$, so the error of the estimate $c_G(\widehat{S})$ is at most the total weighted degree of these vertices, their *volume* $\mathrm{vol}(S \triangle \widehat{S}) = \sum_{v \in S \triangle \widehat{S}} d_v$. We therefore want a partition, fixed before any cut is seen, such that every S can be rounded with $\mathrm{vol}(S \triangle \widehat{S})$ at most a bounded multiple of $c_G(S)$; this multiple becomes the multiplicative error A. The rounding is then built into the explicit graph: J_{aux} has one added vertex per part, labeling the added vertices chooses $\widehat{S}$, and minimizing over their labels rounds S automatically (Step 4 below). The question is what a label change should cost. Charging it to the full weighted degree of a vertex of $S \triangle \widehat{S}$ is prohibitively expensive: if both endpoints of a learned edge of weight M lie in $S \triangle \widehat{S}$, they contribute $2M$ to $\mathrm{vol}(S \triangle \widehat{S})$ although the edge is cut by $\widehat{S}$ exactly when it is cut by S, and M may be far larger than $c_G(S)$. We therefore learn a subgraph K of heavy edges exactly, keep it on the original vertices, and charge label changes only to degrees in the residual graph $R = G - K$, writing r_v for the weighted degree of v in R and $\mathrm{vol}_R(X) = \sum_{v \in X} r_v$. The target is that every shore S admits a union $\widehat{S}$ of parts with

$$\mathrm{vol}_R(S \triangle \widehat{S}) \le k(1+a)c_G(S) + Nt, \tag{15}$$

where $a = O(\log(2N)\log\log(2N))$ is the offline hierarchy factor and t is a *residual degree floor*: vertices of residual degree at most t are simply charged their degree, which contributes at most Nt in total. Choosing $t = \Lambda/(16N^2)$ makes $2Nt$ equal to the additive slack $\varepsilon = \Lambda/(8N)$ allowed in (13). The four steps below first shape the residual so that such a partition exists, then build it, then assemble J_{aux}.

Step 1: learn heavy edges so the residual is spread. We learn an exact subgraph K so that, in the residual graph $R = G - K$, every vertex either has residual weighted degree $r_v \le t$ or has every incident residual edge of weight at most r_v/k. Thus a vertex with substantial residual degree has its remaining weight spread across at least k neighbors. Whenever $r_v > t$, find the edges heavier than a $1/(2k)$ fraction of its current residual degree and move them into K. Rescan only after that degree halves. Normalization bounds the ratio between the initial degree and t by $O(N^3)$, so there are $O(\log N)$ scans per vertex, regardless of the smallest positive edge weight. The cost

is $O(Nk\log^2(2N))$, and afterwards the *row* of every vertex v, the list of residual weights $w_R(v,u)$ over all u, satisfies

$$r_v \le t \quad \text{or} \quad w_R(v,u) \le r_v/k \quad \text{for every } u.$$

Queries to R are simulated by subtracting the known contribution of K from queries to G.

Step 2: learn the rows of a small center set. Next build a center set C of size $O((N/k)\log N)$ with the following property: every noncenter with residual degree above t sends at least r_v/k of its residual weight to C. The row of each center is learned as it is chosen, at $O(N)$ queries per row. This is possible because of Step 1: no single neighbor of such a vertex carries more than a $1/k$ fraction of its residual weight, so a center set covering a $1/k$ fraction of every such row exists, and a greedy argument finds one of the stated size. It can be found deterministically with $O((N^2/k)\log(2N))$ queries. Each vertex v not yet covered carries a weight $p_v = \prod_{c\in C}(1 - kw_R(v,c)/(2r_v))$, which stays above $1/2$ while v is uncovered, and a candidate center u scores

$$\sigma(u) = \frac{k}{2} \sum_{v \text{ uncovered}} \frac{p_v}{r_v}\, w_R(v,u).$$

The scores sum to at least $(k/4)\sum_v p_v$, so some center decreases the potential $\sum_v p_v$ by an $\Omega(k/N)$ fraction. To find one with cut queries, round the coefficients p_v/r_v down to powers of two. The degree floor bounds their ratio by a polynomial in N, so there are $O(\log N)$ buckets A_j with common coefficient θ_j, and the total score of a candidate group W is, up to a factor 2, $\sum_j \theta_j \sum_{v\in A_j}\sum_{u\in W} w_R(v,u)$. Each inner double sum costs $O(1)$ cut queries, even when A_j and W overlap (Lemma 38), and binary search over candidate groups finds a good center. Thus center selection uses only cut queries, without knowing the remaining individual edge weights. This step is a deterministic weighted analogue of the choice of centers in star contraction [AEG+22].

Step 3: a rounding partition that moves little residual volume. Let H_C^R contain the residual edges incident to C, all now known. Using the offline construction of Räcke and Shah, build an a-tree cut sparsifier of the explicit graph $K + H_C^R$, where $a = O(\log(2N)\log\log(2N))$. A tree can be compressed around a terminal set while preserving every minimum extension of its labels: prune branches without terminals, and on each terminal-free degree-two path retain only a minimum-capacity edge. Apply this with the centers as terminals, adding one arbitrary vertex so that the terminal set is nonempty. The compression sends every vertex of G to a node of the compressed tree, and the vertices sent to the same node form a part. This is the rounding partition: it is fixed before any cut is seen, and it has one part per node of the compressed tree, hence $O(1 + (N/k)\log N)$ parts.

Every cut S can be rounded to a union $\widehat{S}$ of these parts without changing its center labels, with $c_{K+H_C^R}(\widehat{S}) \le a\, c_{K+H_C^R}(S)$. For the changed set $X = S \triangle \widehat{S}$, center coverage gives the target bound (15). Indeed, a changed vertex with residual degree above t charges its residual degree to its edges to centers, which carry at least a $1/k$ fraction of it. Since no center changes labels, these edges cross S or $\widehat{S}$. The remaining changed vertices contribute at most Nt.

Step 4: assemble the auxiliary graph. As announced, K stays on the original vertices and only the residual is rounded. Learn the quotient of the whole graph G on the partition of Step 3 with $O(p^2)$ queries, p the number of parts, and place it on one new vertex per part. Join each original vertex to its part vertex with capacity r_v. Keep K on the original vertices. For fixed original

labels S, the minimum extension cost in this explicit auxiliary graph J_{aux} is

$$\Psi(S) = c_K(S) + \min_{\widehat{S} \text{ a union of parts}} \big(c_G(\widehat{S}) + \text{vol}_R(S \triangle \widehat{S})\big). \tag{16}$$

Figure 3 separates the three contributions.

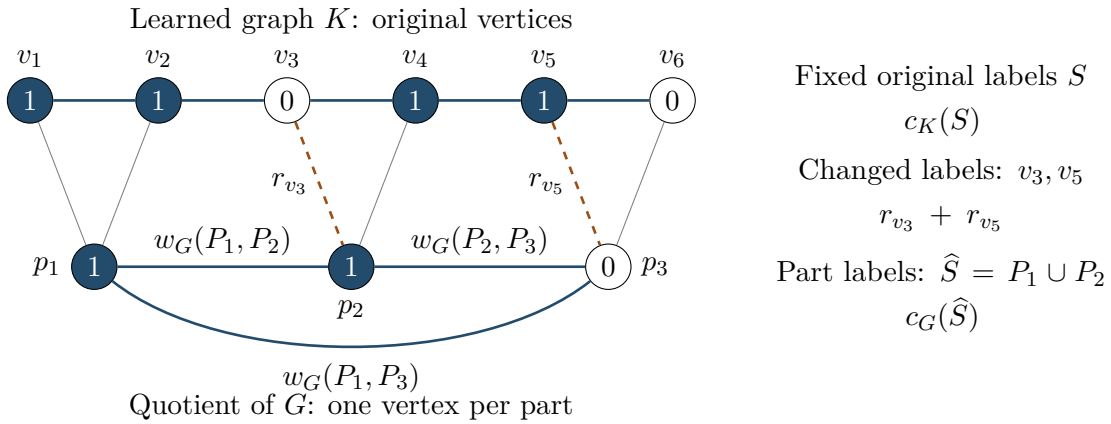


Figure 3: The auxiliary graph for parts $P_1 = \{v_1, v_2\}$, $P_2 = \{v_3, v_4\}$, and $P_3 = \{v_5, v_6\}$, with quotient vertices p_1, p_2, p_3. Each connector has capacity r_v; dashed connectors join unequal labels. The displayed cut has total cost $c_K(S) + c_G(\widehat{S}) + r_{v_3} + r_{v_5}$. Minimizing over part labels gives $\Psi(S)$. Thus only residual weighted degrees pay for label changes, while K stays on the original vertices.

We verify (13). For the lower bound, fix any union of parts $\widehat{S}$. The explicit $c_K(S)$ term accounts for all learned edges, and changing the labels of $S \triangle \widehat{S}$ changes an R-cut by at most the residual volume moved, so $c_R(S) \le c_R(\widehat{S}) + \text{vol}_R(S \triangle \widehat{S}) \le c_G(\widehat{S}) + \text{vol}_R(S \triangle \widehat{S})$. Adding $c_K(S)$ gives $c_G(S) \le \Psi(S)$. For the upper bound, take $\widehat{S}$ to be the rounded cut of Step 3 and write $X = S \triangle \widehat{S}$, so that $\Psi(S) \le c_K(S) + c_G(\widehat{S}) + \text{vol}_R(X)$, and bound the middle term by the rounding guarantee of Step 3 and the same volume argument:

$$c_G(\widehat{S}) = c_K(\widehat{S}) + c_R(\widehat{S}) \le a\, c_G(S) + c_R(S) + \text{vol}_R(X).$$

Hence $\Psi(S) \le (1+a)c_G(S) + 2\,\text{vol}_R(X)$, and (15) gives $A = (1+a)(2k+1) = O(k \log(2N) \log\log(2N))$ and $\varepsilon = 2Nt$. Unlike the contraction classes of normalization, which replace the input by a smaller quotient and must preserve every minimum cut, the rounding partition need not be safe: it is used only to construct J_{aux}, and Ψ optimizes over its labels separately for each cut.

5.4 The query bound

Write $L = \lceil \log_2(2N) \rceil$ and $K = \lceil \log_2(2L) \rceil$, and choose an absolute constant D such that a successful phase has quality at most DkL^2K^2. Set $k = \lfloor \alpha/(DL^2K^2) \rfloor$; the lower bound on α ensures $k \ge 2$. If $k \ge N$, reconstruct the current quotient and compute its offline hierarchy, whose quality is at most α. Otherwise the phase of Section 5.2 either succeeds with quality at most α or halves N.

The rounding partition has $p \le N$ parts and $p = O((N/k)L)$, so quotient reconstruction costs $O(p^2) = O(N^2L/k)$ queries. Including the residual preparation, center rows, and the test, a phase uses

$$O\left(NkL^2 + \frac{N^2L}{k}\right) = O\left(N\alpha + \frac{N^2L^3K^2}{\alpha}\right)$$

queries. A final direct reconstruction costs at most $O(N\alpha)$. The phase sizes decrease geometrically, and all normalization calls together use $O(n \log(2n))$ queries. Summing gives $O(n\alpha + (n^2/\alpha)\log^3(2n)(\log\log(2n))^2)$, which is $O((n^2/\alpha)\log^3(2n)(\log\log(2n))^2)$ because $\alpha^2 \le n$. The contracted classes form a safe partition, proving the claimed guarantees of Theorem 7. Combined with Theorem 6, this completes the overview of Theorem 4; the lift of the quotient skeleton back to $V(G)$ and the final two-respecting search are in Section 11.8.

6 Preliminaries and query primitives

An algorithm receives the vertex set and the exact cut oracle, and only oracle calls are charged. Computation on known quantities is free, including exhaustive search and linear programming on explicit graphs, and arithmetic is exact. Weights are arbitrary nonnegative reals, with no assumption on their magnitudes or ratios; parallel edges are aggregated and loops ignored, and in the unweighted case the input is a simple graph. A certificate may be chosen with full knowledge of G but must be sound for every nonnegative graph agreeing with the queried values, and its size is its number of queries. All algorithms are deterministic, with fixed tie-breaking, and always correct.

We write $d_v = c_G(\{v\})$ for the weighted degree of v, the sum of its incident edge weights, and $\mathrm{vol}_G(X) = \sum_{v\in X} d_v$. For disjoint sets A, B,

$$2w_G(A,B) = c_G(A) + c_G(B) - c_G(A \cup B). \tag{17}$$

Thus an individual edge weight or a partial-star total costs at most three cut queries. Once the weighted degrees are known, a row mass costs at most two further queries:

$$2w_G(v,X) = d_v + c_G(X) - c_G(X \cup \{v\}) \qquad (v \notin X). \tag{18}$$

For general X, remove v before applying this formula. The values of empty and full cuts are known to be zero.

Two other elementary inequalities will be used repeatedly:

$$c_G(S \triangle T) \le c_G(S) + c_G(T), \qquad |c_G(S) - c_G(T)| \le \mathrm{vol}_G(S \triangle T). \tag{19}$$

Both follow by checking each nonnegative edge contribution. If an explicit subgraph K with original edge weights has been learned, cut queries to $G - K$ are simulated by subtracting $c_K(S)$ from $c_G(S)$. A query to a contracted graph is one original query on a union of its classes.

Lemma 8 (Relative-heavy row search)**.** *Suppose a residual row has known total mass $M > 0$ and queryable nonnegative group masses. For $b \ge 1$, all its entries exceeding M/b can be found, with exact values, using $O(b\log(2n))$ cut queries.*

Proof. Arrange the possible neighbors in a balanced binary tree. Explore the children of a group only when its mass exceeds M/b. At each depth fewer than b disjoint groups pass this test. Their children cost $O(b)$ row-mass queries, each implemented by (18). Every entry above the threshold is reached at a leaf, and there are $O(\log(2n))$ depths. A row of mass zero needs no search. □

Lemma 9 (Positive-support forest)**.** *A maximal spanning forest of the positive support can be learned deterministically in $O(n\log(2n))$ cut queries.*

Proof. Maintain the components of a partial forest. Query the cut of a component A. If it is positive, find a positive edge from A to its complement by two balanced searches using (17): first locate its

endpoint in A, then its endpoint outside A. Learn the weight and merge the two components. A component with zero cut is finished and remains finished. Initially there are n components, each merge creates one new component, and each component is tested at most once before being merged or finished. There are $O(n)$ tests and at most $n-1$ edge searches. □

In particular, disconnected positive support yields a zero optimum within this query bound. All skeleton constructions below use $\lambda = \lambda(G) > 0$. The graph Laplacian is denoted by L_G, so that $c_G(S) = \mathbf{1}_S^{\mathsf{T}} L_G \mathbf{1}_S$.

7 Near-linear certificates

A queried optimum shore supplies the upper bound. We build the matching lower bound from feasible divergences: first for one terminal problem, then for all isolating cuts of a terminal set, and finally for every nontrivial cut by halving that set. Every query in this section may depend on the entire input graph.

7.1 Certifying flows and isolating cuts

Label the vertices $0, \dots, n-1$, orient each pair from its smaller to its larger endpoint, and let D be the incidence matrix, with the convention that Df is net outflow. We also write a signed flow antisymmetrically as $f_{uv} = -f_{vu}$, with $f_{uv} > 0$ meaning flow from u to v. Feasibility means $|f_{uv}| \le w_{uv}$.

Proof of Lemma 2. Fixed costs and a minimum-cost flow. For $u < v$ give the pair uv the auxiliary cost $a_{uv} = 2^{un+v}$, and put $a_{vu} = -a_{uv}$. These costs are fixed before G and b. Every nonempty signed sum of distinct pair costs is nonzero: its largest power of two exceeds the sum of all smaller powers. In particular, the signed cost around any simple undirected cycle is nonzero.

Choose a flow f minimizing

$$\sum_{u<v} a_{uv} f_{uv} \qquad \text{subject to} \qquad Df = b, \quad -w \le f \le w.$$

A minimizer exists because the feasible polytope is nonempty and compact. A residual arc $u \to v$ is available whenever $f_{uv} < w_{uv}$ and has cost a_{uv}. There is no negative-cost residual cycle, since sending a sufficiently small circulation around one would decrease the objective while preserving feasibility and divergence.

Attach a new source with zero-cost arcs to all vertices and let ℓ_v be the shortest-path distance to v in this residual graph. These distances exist: removing nonnegative-cost closed subwalks leaves a simple path, and there are only finitely many simple paths. Set $p_v = -\ell_v$ and define the reduced costs

$$\rho_p(u, v) = a_{uv} - p_u + p_v.$$

Every residual arc has nonnegative reduced cost. Applying this to both orientations gives the standard complementary-slackness conditions (see Bertsekas [Ber91, Section 1.2]):

$$\rho_p(u,v) < 0 \implies f_{uv} = w_{uv}, \qquad \rho_p(u,v) > 0 \implies f_{uv} = -w_{uv}. \tag{20}$$

A tree determines the saturation directions. Consider the graph of all pairs with zero reduced cost, including pairs of capacity zero. It is a forest: a cycle of zero reduced-cost pairs would have zero signed cost, since the potentials telescope. If this forest is disconnected, let A be one of its

components. Among the ordered pairs $u \in A$, $v \notin A$, choose one minimizing $|\rho_p(u,v)|$, and put $\tau = \rho_p(u,v) \neq 0$. Add τ to every potential in A. Reduced costs within A or its complement are unchanged; a reduced cost directed out of A decreases by τ. By the choice of $|\tau|$, no nonzero reduced cost changes to the opposite sign, and the chosen pair becomes zero. All old zero pairs remain zero, so the zero forest strictly grows. Equation (20) still holds for the same flow. Repeating produces a spanning tree T of zero reduced-cost pairs.

Normalize the final potential to be zero at vertex 0. It is now the unique potential p^T satisfying

$$p_0^T = 0, \qquad p_u^T - p_v^T = a_{uv} \quad (uv \in T),$$

obtained by summing the fixed costs along tree paths. Thus p^T depends only on T. Write $\rho_T = \rho_{p^T}$. Its zero pairs are exactly the tree edges, and (20) specifies the flow on every non-tree pair:

$$f_{uv} = -\operatorname{sgn}(\rho_T(u,v))\, w_{uv} \qquad (uv \notin T). \tag{21}$$

The tree need not lie in the positive support of G, and some of its edges may themselves be saturated.

The queries and their verification. For each vertex v, the tree determines the outgoing non-tree set

$$O_T(v) = \{u \neq v : uv \notin T,\ \rho_T(v,u) < 0\}.$$

Query all singleton shores, all two-vertex shores $\{u,v\}$ with $uv \in T$, and the two shores $O_T(v)$ and $O_T(v) \cup \{v\}$ for each v. This is at most $n + (n-1) + 2n = 4n - 1$ cut queries. By (17), the singleton and two-vertex answers determine every tree capacity. The divergence b_v^{nt} of the prescribed non-tree flow is twice its outgoing capacity minus its total incident capacity. Substituting (18) gives

$$b_v^{\mathrm{nt}} = c_G(O_T(v)) - c_G(O_T(v) \cup \{v\}) + \sum_{u:uv \in T} w_{uv}, \tag{22}$$

which is determined by the answers.

In a consistent graph G', retain f on the tree and use (21) with the new capacities on every non-tree pair. Tree capacities are unchanged, so this flow is feasible. Equation (22) preserves its non-tree divergence, and hence its full divergence is still b. This also handles new positive edges: their prescribed directions were already determined by T.

The verifier does not need the original tree-edge flow values. Given b and the answers, its tree flow is the unique one with divergence $b - b^{\mathrm{nt}}$. Explicitly, if deleting uv from T leaves the component A containing u, then its value from u to v is

$$\sum_{x \in A} (b_x - b_x^{\mathrm{nt}}).$$

Check that the residual divergences sum to zero and that each of these $n-1$ values respects its measured tree capacity. These checks are sufficient for feasibility, as follows by successively removing leaves of the tree. Thus T fixes all queried shores and all non-tree directions; b and the answers determine the tree flow.

For distinct s, t, take the divergence of a maximum s–t flow and add a query for a matching minimum shore S^*. Its answer determines $b_s = c_G(S^*)$, $b_t = -c_G(S^*)$, and all other entries zero. The feasible divergence supplies the lower bound in every consistent graph, and the queried shore supplies the equal upper bound. The certificate is described by (T, S^*) and uses at most $4n$ queries.

For the terminal-set assertion, add a source joined to every vertex of A and a sink joined to every vertex of B, each added edge with a fixed capacity greater than $\sum_e w_e$. A maximum flow in this augmented network has value ν equal to the minimum A–B separating-cut value in G. Its

restriction to the original edges is a feasible flow in G whose divergence b is nonnegative on A, nonpositive on B, zero elsewhere, and sums to ν over A. Certify b by the construction above, with at most $4n-1$ queries to G. In every consistent graph b admits a feasible flow, so every shore S with $A \subseteq S$ and $S \cap B = \emptyset$ satisfies $c(S) \geq \sum_{v \in S} b_v = \nu$. Query one original minimum A–B cut for the matching upper bound. Alternatively, the construction can be run on the augmented network itself, whose added capacities are constants; each of its cut queries is then a bounded number of cut queries to G (Remark 10). □

The certified object is the divergence b, not the chosen flow f. The proof constructs a possibly different capacity-respecting flow for each consistent graph. This is exactly the lower-bound witness needed for terminal cuts, even when the competitor adds new edges.

Remark 10. The same argument applies to a network on a subset $W \subseteq V$ whose unknown edges are the original edges within W, together with a constant number of auxiliary vertices and fixed-capacity edges. It uses $O(|W|+1)$ cut queries on G: the unknown part of each local cut is $w_G(A, W \setminus A)$ for some $A \subseteq W$, which takes at most three original cut queries by (17); the fixed contributions are added exactly. The fixed capacities need not dominate the capacities of a competitor.

For a terminal set $R \subseteq V$, $|R| \geq 2$, and $v \in R$, put

$$\mathrm{iso}_G(v; R) = \min\{c_G(S) : S \cap R = \{v\}\}.$$

Lemma 11 (Isolating-cut certificates)**.** *For every $R \subseteq V$ with $|R| \geq 2$, all the values $\mathrm{iso}_G(v; R)$, $v \in R$, can be certified simultaneously with $O(n \log |R|)$ cut queries.*

Proof. Use the isolating-cut construction of Li and Panigrahi [LP25]. Give the terminals distinct binary labels of length $k = \lceil \log_2 |R| \rceil$. For each bit, choose a minimum cut separating the terminals with bit zero from the terminals with bit one. Certify that cut's optimality by Lemma 2, using $O(nk)$ queries in total. Each chosen cut is therefore still optimal for its terminal partition in every consistent graph.

For $v \in R$, let P_v be the intersection of the chosen bit-cut shores containing v. The cells P_v are disjoint and satisfy $P_v \cap R = \{v\}$. An inclusion-minimal minimum isolating shore for v lies in P_v. Indeed, if such a shore S is not contained in a bit-cut shore T containing v, submodularity gives

$$c_G(S \cap T) + c_G(S \cup T) \leq c_G(S) + c_G(T).$$

The intersection still isolates v, and the union still separates the two terminal classes for this bit. Their optimality inequalities force equality, making $S \cap T$ a strictly smaller minimum isolating shore, a contradiction. The same argument applies in every consistent graph because the bit cuts remain optimal there.

For each $v \in R$, let G_v be the graph obtained from G by contracting $V \setminus P_v$ to a single vertex t_v, and certify a minimum v–t_v cut of G_v by Lemma 2, using at most $4(|P_v|+1)$ queries. Each cut query to G_v is one cut query to G: a shore $X \subseteq P_v$ has value $c_{G_v}(X) = c_G(X)$, and a shore containing t_v has value $c_G(P_v \setminus (X \setminus \{t_v\}))$. A graph consistent with these queries on G contracts to a graph consistent with them on G_v, so the certificate remains valid there: in every consistent graph, every shore $S \subseteq P_v$ with $v \in S$ has value at least the certified minimum. Since the cells are disjoint, the certificates of all the contracted graphs together cost

$$O\Big(\sum_{v \in R} (|P_v|+1)\Big) = O(n).$$

The containment property gives the required lower bounds on the global isolating-cut values. For the upper bounds, query one original minimum isolating shore for each terminal. Together with the bit-cut certificates, this uses $O(n \log |R|)$ queries.

Alternatively, the local network can be formed explicitly on $P_v \cup \{t_v\}$: query the boundary capacity $d_v^{\text{out}}(u) = w_G(\{u\}, V \setminus P_v)$ of each $u \in P_v$, at most n partial-star queries in all, and join u to t_v by an edge of this fixed capacity. The capacity of every shore $S \subseteq P_v$ in this network equals $c_G(S)$, also in every consistent graph, since the boundary totals were fixed, and the flow certificate of Remark 10 again costs $O(|P_v| + 1)$ queries. □

7.2 Halving the terminal set

Throughout this subsection put $\lambda = \lambda(G) > 0$. All flows use the original vertex set, even when the terminal set becomes smaller.

Call $S \subseteq V$ *U-separating* if $S \cap U \neq \emptyset$ and $U \not\subseteq S$. For a symmetric weighting h write c_h for its cut function.

Lemma 12 (Fractional splitting off)**.** *Let $U \subseteq V$ and suppose $c_G(S) \geq \lambda$ for every U-separating S. There is a nonnegative weighted graph J on V, with weight zero on every pair having an endpoint outside U, such that*

$$c_J(S) \leq c_G(S) \quad \text{for every } S, \qquad c_J(S) \geq \lambda \quad \text{for every } U\text{-separating } S.$$

Proof. Let $\mathcal{H}$ be the set of symmetric nonnegative weightings h on V with zero diagonal satisfying both displayed families of inequalities. It contains w_G and is closed. It is bounded, since $h(x, y) \leq c_h(\{x\}) \leq c_G(\{x\})$. Choose $h \in \mathcal{H}$ minimizing the weight $\Phi(h) = \sum_{z \notin U} \sum_y h(z, y)$ at nonterminals, and let H be the corresponding graph. Call a pair with positive H-weight an edge, and call a U-separating set *tight* if its H-cut equals λ. Complements of tight sets are tight.

Suppose, for a contradiction, that some nonterminal x has an edge. Let D be a symmetric weighting with zero diagonal that is negative only on edges, with $c_D(S) \leq 0$ for every S and $\Phi(D) < 0$. If no tight set S had $c_D(S) < 0$, then $h + \varepsilon D \in \mathcal{H}$ for all small $\varepsilon > 0$, because there are finitely many sets; this would contradict minimality. We use two such directions.

(i) Deleting an edge xu: $c_D(S)$ is -1 if S separates x from u and 0 otherwise, and $\Phi(D) \leq -1$. Hence, after taking a complement, some tight set contains u but not x.

(ii) Splitting distinct edges xu, xv into uv: then $c_D(S) \in \{0, -2\}$, with $c_D(S) = -2$ exactly when S separates x from both u and v, and $\Phi(D) = -2$ (the weight at x drops by 2, while at u and at v the lost edge to x is offset by the new edge uv, whether or not they are terminals). Hence some tight set contains u and v but not x.

Among the tight sets that avoid x and contain a neighbor of x, choose X of maximum size, containing the neighbor u. The set $X \cup \{x\}$ is U-separating because $x \notin U$, so

$$\lambda \leq c_H(X \cup \{x\}) = \lambda + d_H(x) - 2w_H(x, X).$$

Thus $w_H(x, X) \leq d_H(x)/2$. Since $w_H(x, X) \geq H(x, u) > 0$, some neighbor v of x lies outside X. By (ii) there is a tight set Y avoiding x and containing u and v.

If $X \cap Y$ and $X \cup Y$ are both U-separating, submodularity gives $c_H(X \cup Y) \leq 2\lambda - c_H(X \cap Y) \leq \lambda$. Then $X \cup Y$ is a tight set avoiding x, containing u, and larger than X because it contains v, a contradiction. Otherwise either $X \cap Y$ contains no terminal or $X \cup Y$ contains all terminals. In

both cases $X \setminus Y$ contains a terminal outside Y and misses the terminals of Y, so it is U-separating, and symmetrically so is $Y \setminus X$. The identity

$$c_H(X) + c_H(Y) = c_H(X \setminus Y) + c_H(Y \setminus X) + 2w_H(X \cap Y, V \setminus (X \cup Y))$$

then gives $w_H(X \cap Y, V \setminus (X \cup Y)) \le 0$, contradicting $u \in X \cap Y$, $x \notin X \cup Y$, and $H(u,x) > 0$. Hence no nonterminal has an edge, and $J = H$ satisfies the lemma. □

Lemma 13 (A flow to half the terminals)**.** *For every $U \subseteq V$ with $|U| \ge 2$, there is a nonempty $R \subset U$ with $|R| \le |U|/2$ and a feasible flow whose divergence satisfies*

$$b_u = \lambda/2 \quad (u \in U \setminus R), \qquad b_v = 0 \quad (v \notin U), \qquad b_v \le 0 \quad (v \in R).$$

Consequently, $O(n)$ cut queries certify the inequalities

$$c_{G'}(S) \ge \frac{\lambda}{2}|S \cap U| \qquad \text{for every } S \text{ with } S \cap R = \emptyset. \tag{23}$$

Proof. Every U-separating set is a nontrivial cut, so Lemma 12 applies and gives J. For $u \in U$ the singleton $\{u\}$ is U-separating because $|U| \ge 2$, so $d_J(u) \ge \lambda$. Since J has no weight outside U, $c_J(S) = c_J(S \cap U)$ for every S; in particular $c_J(U \setminus X) = c_J(X)$ for $X \subseteq U$.

Choose $X \subseteq U$ maximizing $c_J(X)$, and let R be the smaller of X and $U \setminus X$. Then $|R| \le |U|/2$, and R also maximizes c_J. For $u \in U \setminus R$, moving u into R cannot increase the cut:

$$c_J(R \cup \{u\}) = c_J(R) + d_J(u) - 2w_J(u, R) \le c_J(R).$$

Hence $w_J(u, R) \ge d_J(u)/2 \ge \lambda/2$. Also $c_J(R) \ge c_J(\{u\}) \ge \lambda > 0$ for any $u \in U$, so R is neither empty nor all of U. For every S with $S \cap R = \emptyset$,

$$c_G(S) \ge c_J(S) \ge w_J(S \cap U, R) \ge \frac{\lambda}{2}|S \cap U|. \tag{24}$$

Add a source s joined to every $u \in U \setminus R$ with capacity $\lambda/2$, and a sink t joined to every vertex of R with a fixed capacity $M > (\lambda/2)|U \setminus R|$. Any cut containing a vertex of R on its source side already has capacity at least M. Every remaining source–sink cut has source side $\{s\} \cup S$ with $S \cap R = \emptyset$, and capacity

$$c_G(S) + \frac{\lambda}{2}|(U \setminus R) \setminus S| \ge \frac{\lambda}{2}|(U \setminus R) \cap S| + \frac{\lambda}{2}|(U \setminus R) \setminus S| = \frac{\lambda}{2}|U \setminus R|$$

by (24). By max-flow/min-cut, a maximum flow saturates every source edge. Its restriction to G is a feasible flow with the required divergences.

Finally certify this fixed divergence vector by Lemma 2. A shore disjoint from R has total divergence $(\lambda/2)|S \cap U|$, giving (23) in every consistent graph. □

Proof of Theorem 1. If $\lambda(G) = 0$, one zero-weight nontrivial cut suffices. Otherwise start with $U_0 = V$. At every stage with $|U_i| \ge 2$, certify all isolating-cut values for U_i by Lemma 11, and use Lemma 13 to choose and certify $U_{i+1} \subset U_i$ with $1 \le |U_{i+1}| \le |U_i|/2$. Stop when U_L is a singleton. Also query one minimum cut of G. There are at most $\lceil \log_2 n \rceil$ rounds, and the total number of queries is

$$O\left(n \sum_{i=0}^{L-1} \log(2|U_i|)\right) = O(n \log^2(2n)).$$

Suppose that a consistent graph has a nontrivial cut S of capacity less than λ. Both shores meet U_0. If both meet U_i, neither contains exactly one terminal $v \in U_i$, since it would then have capacity at least

$$\mathrm{iso}_{G'}(v; U_i) = \mathrm{iso}_G(v; U_i) \geq \lambda.$$

Thus both shores contain at least two terminals of U_i. By (23), neither can be disjoint from U_{i+1}. Induction implies that both meet the singleton U_L, a contradiction. Hence every nontrivial cut of G' has capacity at least λ. The queried minimum cut gives the reverse inequality. □

The split-off graphs and the maximum cuts used to choose retained terminals are selected with knowledge of G. Only the resulting divergence vectors are certified; the graphs J need not remain valid for a consistent graph. Lemma 12 is a fractional form of the splitting-off theorems of Lovász [Lov76] and Mader [Mad78]: it preserves only connectivity across the terminal set, which is all the maximum-cut argument uses, and it needs neither integrality nor the preservation of pairwise connectivities. The proof above works directly with arbitrary real capacities and requires no rational approximation or integrality hypothesis.

8 Weak skeletons and tree covers

We now pass from certifying an optimum to an explicit graph in which it can be found. The certificate construction gives an exact completion after its measurements are supplied. For an algorithm choosing its own measurements, the weaker condition below suffices: a short tree cover of the weak skeleton reduces the final search to $O(n \log^2 n)$ further queries to the original graph.

Recall from Definition 3 that, for a graph G with $\lambda = \lambda(G) > 0$, a weak skeleton is an explicit nonnegative weighted graph H on $V(G)$ with

$$\lambda(H) \geq \lambda, \qquad c_H(S^*) \leq \tfrac{5}{4}\lambda \quad \text{for every minimum-cut shore } S^* \text{ of } G. \tag{25}$$

Thus the positive support of H is connected. Edges absent from G may be used in H. The tree cover below uses only that every minimum cut of G is a 5/4-approximate minimum cut of H.

Remark 14 (Sparse output). The number of edges of H never enters a query bound. If an explicit graph H_0 satisfies $\lambda(H_0) \geq \lambda$ and $c_{H_0}(S^*) \leq \frac{121}{100}\lambda$ for every minimum cut S^*, deterministic spectral sparsification [BSS12], scaled so that $L_{H_0} \preceq L_H \preceq \frac{101}{100}L_{H_0}$, gives an $O(n)$-edge weak skeleton H without further queries, since $c_H(S^*) \leq (12221/10000)\lambda < (5/4)\lambda$.

8.1 Certificates give sparse consistent skeletons

Proposition 15 (A certificate-to-skeleton connection)**.** *Let $\mathcal{Q}$ be a q-query minimum-cut certificate for G that includes an optimum shore S^*. There is a nonnegative graph F with at most q edges, consistent with every answer, such that*

$$\lambda(F) = c_F(S^*) = \lambda(G).$$

Given the certificate shores and answers, such an F can be computed without further queries. If $\lambda(G) > 0$, then F satisfies (25) for the certified shore S^*, which is all the tree cover of Lemma 17 needs to find S^*; F need not approximate the other minimum cuts of G.

Proof. Let A have the queried cut-incidence vectors as rows, and let $a = Aw_G$. The system $Ax = a$, $x \geq 0$, is feasible. Choose a feasible solution with minimal support. If its supported columns were linearly dependent, moving along a nonzero dependence in one of its two directions until a positive

coordinate became zero would preserve feasibility and reduce support. Thus its support has size at most $\operatorname{rank}(A) \le q$. This solution can be found by linear programming, or by enumerating supports and solving their linear feasibility problems, using only A, a.

Every feasible solution has minimum-cut value $\lambda(G)$ by the certificate property. Since S^* was queried, its value is also fixed to $\lambda(G)$. This proves the last claim. The support of the computed completion need not be known to be contained in the support of G. □

The certificate in Theorem 1 already includes S^*. It therefore gives a graph-dependent collection of $O(n \log^2 n)$ measurements sufficient for constructing an explicit graph in which the certified minimum cut can be found. The algorithmic question is how to select measurements from oracle answers alone. The next sections answer this by constructing a weak skeleton directly. They do not invert the certificate proof or recover its terminal sets and flow witnesses.

8.2 Covering near-minimum cuts by a few trees

We use the minimum-spanning-tree inequality behind fractional tree packing, in the following elementary form. For a set E of pairs and a cut S, write $\delta_E(S)$ for the pairs of E with one endpoint in S. Call E *spanning* if $\delta_E(S) \neq \emptyset$ for every nontrivial cut S.

Lemma 16 (Light spanning edge sets)**.** *Let H be a nonnegative weighted graph on V with $\lambda(H) \ge \lambda > 0$, and let $\ell \ge 0$ be lengths on the pairs of V. There is a spanning set E of positive-weight pairs of H with*

$$\sum_{e \in E} \ell_e \le \frac{2}{\lambda} \sum_{\{a,b\}} w_H(a,b)\, \ell_{ab}. \tag{26}$$

Proof. We prove a statement for partitions. Let $\mathcal{P}$ be a partition of V into $k \ge 1$ blocks such that every nonempty proper union of blocks has H-cut at least λ. Say that E *connects* $\mathcal{P}$ if every such union is crossed by a pair of E. We show that, if ℓ is nonnegative on the positive-weight pairs crossing $\mathcal{P}$, there is a set E of at most $k-1$ positive-weight pairs connecting $\mathcal{P}$ with

$$\sum_{e \in E} \ell_e \le \frac{2}{\lambda} \sum_{\{a,b\} \text{ crossing } \mathcal{P}} w_H(a,b)\, \ell_{ab}.$$

The lemma is the case of singleton blocks.

For $k = 1$ take $E = \emptyset$. Let $k \ge 2$. A single block is a proper union of blocks, so its cut is at least $\lambda > 0$ and some positive-weight pair crosses $\mathcal{P}$. Among these, choose uv with the least length $\ell_0 = \ell_{uv} \ge 0$. Merge the blocks of u and v to obtain $\mathcal{P}'$ with $k-1$ blocks. Unions of blocks of $\mathcal{P}'$ are unions of blocks of $\mathcal{P}$, and the lengths $\ell' = \ell - \ell_0$ are nonnegative on the positive-weight pairs crossing $\mathcal{P}'$, since those pairs also cross $\mathcal{P}$. By induction there is a set E' of at most $k-2$ such pairs connecting $\mathcal{P}'$, with the bound for ℓ'. Put $E = E' \cup \{uv\}$. A union of blocks of $\mathcal{P}$ that does not separate u from v is a union of blocks of $\mathcal{P}'$ and is crossed by E'; otherwise it is crossed by uv. Moreover

$$\sum_{e \in E} \ell_e \le \ell_0 + \sum_{e \in E'} (\ell'_e + \ell_0) \le \sum_{e \in E'} \ell'_e + (k-1)\ell_0.$$

The total weight W of the pairs crossing $\mathcal{P}$ is half the sum of the k block cuts, so $W \ge k\lambda/2$. Pairs that cross $\mathcal{P}$ but not $\mathcal{P}'$ have $w_H \ell' \ge 0$. Hence

$$\frac{2}{\lambda} \sum_{\text{crossing } \mathcal{P}} w_H \ell = \frac{2}{\lambda} \sum_{\text{crossing } \mathcal{P}} w_H \ell' + \frac{2W}{\lambda}\, \ell_0 \ge \frac{2}{\lambda} \sum_{\text{crossing } \mathcal{P}'} w_H \ell' + k\ell_0 \ge \sum_{e \in E} \ell_e. \qquad \square$$

Lemma 17 (A deterministic tree cover)**.** *From an explicit nonnegative weighted graph H with $\lambda(H) > 0$, one can deterministically compute $O(\log n)$ spanning trees on V such that every cut of weight at most $(5/4)\lambda(H)$ crosses at most two edges of at least one tree. No queries to G are required.*

Proof. Put $\lambda = \lambda(H)$ and let $\mathcal{F}$ be the set of nontrivial cuts of weight at most $(5/4)\lambda$, each counted once up to complementation. We first show that for every nonempty $\mathcal{F}' \subseteq \mathcal{F}$ some spanning tree T crosses at most two edges of at least $|\mathcal{F}'|/4$ of its cuts.

Let ℓ_{ab} be the number of cuts in $\mathcal{F}'$ that separate a from b. Then $\sum_{\{a,b\}} w_H(a,b)\ell_{ab} = \sum_{S\in\mathcal{F}'} c_H(S)$, and Lemma 16 gives a spanning set E with

$$\sum_{S\in\mathcal{F}'} |\delta_E(S)| = \sum_{e\in E} \ell_e \le \frac{2}{\lambda} \sum_{S\in\mathcal{F}'} c_H(S) \le \frac{5}{2}\,|\mathcal{F}'|.$$

Each term on the left is at least one because E is spanning. If b cuts have $|\delta_E(S)| \ge 3$, the left side is at least $|\mathcal{F}'| + 2b$, so $b \le \frac{3}{4}|\mathcal{F}'|$. The graph (V, E) is connected; let T be any spanning tree of it, for instance a breadth-first tree. Since $T \subseteq E$, we have $|\delta_T(S)| \le |\delta_E(S)|$, and T crosses at most two edges of at least $|\mathcal{F}'|/4$ cuts of $\mathcal{F}'$.

A fixed spanning tree crosses at most two edges of at most $\binom{n}{2}$ nontrivial cuts, up to complementation: choose one or two of its edges. Applying the first step to $\mathcal{F}' = \mathcal{F}$ therefore gives $|\mathcal{F}| \le 4\binom{n}{2}$. Now choose trees greedily, each time applying the first step to the cuts not yet covered. After t choices at most $(3/4)^t \cdot 4\binom{n}{2}$ cuts remain, so $t = O(\log n)$ trees cover $\mathcal{F}$.

Every step concerns only the explicit graph H, so this is a deterministic computation in the query model. The proof of Lemma 16 is itself a greedy procedure: repeatedly add a shortest positive-weight pair joining two different components. □

Remark 18. The inequality (26) is the minimum-spanning-tree bound that is dual to the fractional spanning-tree packing theorem. Using it directly with lengths that count uncovered cuts avoids computing an optimal packing. All choices in this cover use only the explicit skeleton. The construction has no random choices and does not use the original graph's unqueried weights.

Theorem 19 (Two-respecting cuts)**.** *Given an explicit tree T on $V(G)$, a minimum G-cut crossing at most two edges of T can be found deterministically with $O(n \log n)$ cut queries. The edges of T need not be edges of G.*

Section 12 proves this theorem by implementing the interested-path reduction of Gawrychowski, Mozes, and Weimann [GMW21]. Boundary-measure searches locate all interest paths with $O(n \log n)$ queries, and SMAWK [AKM+87] searches the resulting full Monge rectangles within the same budget.

Corollary 20. *An algorithm constructing a weak skeleton in $q(n)$ queries yields a deterministic minimum-cut algorithm using $q(n) + O(n \log^2 n)$ queries. The same holds for an explicit graph satisfying (25) for a single minimum cut.*

Proof. Compute the tree cover of Lemma 17 on the explicit skeleton and apply Theorem 19 to each tree. Every minimum cut of G satisfying (25) is among the covered cuts, and every returned candidate is an actual cut of G. □

9 A weak skeleton from a supplied hierarchy

The certificate-to-skeleton connection used full input knowledge to select measurements. From this section onward, every query is selected from supplied data and previous answers. We first isolate what a known hierarchy must guarantee about the unknown minimum cuts; Section 11 will construct such a hierarchy.

A hierarchy $\mathcal{T}$ is a rooted tree whose leaves are precisely V and whose internal nodes have at least two children. A node represents its descendant leaf set. We also admit, as the formalization does, unary nodes and leaves that are not in V, provided that the nodes of V have no children. A nonroot node u then represents the cluster $A_u = \mathrm{sub}(u) \cap V$ of terminals below it; distinct nodes may represent the same cluster, and a cluster may be $\emptyset$ or V. Every hierarchy constructed below has at most $2n+1$ nodes, and the hierarchy supplied at the start has $O(n)$ nodes, which affects only the constant in the first round. Write $\mathcal{A}$ for the nonroot clusters, indexed by nodes, so $|\mathcal{A}| = O(n)$. The *canonical* capacities for G are $d_A = c_G(A)$ on the edge from a cluster A to its parent, so $d_A = 0$ when $A \in \{\emptyset, V\}$ and $d_A \geq \lambda$ otherwise; they are learned in $O(n)$ queries. Define the representation cost

$$R(S) = \min_{\substack{\text{binary labels on tree nodes} \\ \text{labels on } V \text{ equal } \mathbf{1}_S}} \sum_{A \in \mathcal{A}} d_A |\ell(A) - \ell(\mathrm{parent}(A))|. \tag{27}$$

It is the minimum cut in the tree extending the prescribed labels on V; leaves outside V are unconstrained.

Theorem 21 (Supplied-hierarchy bound)**.** *Suppose $\lambda(G) > 0$, $\mathcal{T}$ and $B \geq 1$ are supplied, and every minimum cut S^* of G satisfies $R(S^*) \leq B\lambda(G)$. A weak skeleton can be constructed deterministically in $O(nB^2 \log n)$ cut queries. The bound is independent of hierarchy depth, and neither the minimum cuts nor $\lambda(G)$ are needed as input.*

9.1 Quantiles of boundary measures

Lemma 22 (Exact checkpoints for a nonnegative measure)**.** *Let μ be a nonnegative measure on an ordered set $[n]$, with known total mass $M > 0$. Suppose prefix masses can be queried. For an integer $q \geq 1$, $O(q \log n)$ queries give $O(q)$ exact prefix constraints such that every nonnegative measure ν satisfying those constraints obeys $|\mu([x]) - \nu([x])| \leq M/q$ for every prefix $[x]$, $0 \leq x \leq n$. Consequently*

$$|\mu(I) - \nu(I)| \leq \frac{2M}{q} \qquad \text{for every interval } I \subseteq [n]. \tag{28}$$

More precisely, at most $(q-1)(\lfloor \log_2 n \rfloor + 3)$ prefix queries and at most $2q$ recorded prefixes suffice.

Proof. For $j = 1, \ldots, q-1$, find by binary search the first index k_j with

$$\mu([k_j]) \geq jM/q.$$

Record the exact masses of the prefixes ending at $k_j - 1$ and k_j, as well as the masses at indices 0 and n. Put $k_0 = 0$ and $k_q = n+1$; recorded indices need not be distinct.

Fix $0 \leq x \leq n$ and let j be the largest index with $k_j \leq x$. Then $k_j \leq x \leq k_{j+1} - 1$, and both endpoints are recorded (for $j = q-1$ the right endpoint is n). Their μ-masses lie in $[jM/q, (j+1)M/q]$: the left one by the choice of k_j, and the right one because k_{j+1} is the first index reaching $(j+1)M/q$, or because the total is M. Since ν agrees with μ at both endpoints and both measures are nonnegative, $\mu([x])$ and $\nu([x])$ lie in this interval of length M/q. Recording both sides of each crossing is essential when there is a large atom. An interval is the difference of two prefixes, which proves (28). □

Choose a depth-first order of the leaves in V of the supplied hierarchy. Every cluster is an interval in this order. For each $A \in \mathcal{A}$, use the two boundary measures

$$\mu_A^{\mathrm{out}}(X) = w_G(A, X \setminus A),$$
$$\mu_A^{\mathrm{in}}(X) = w_G(A \cap X, V \setminus A).$$

Both have total mass d_A; when $d_A = 0$ both vanish identically and no queries are needed. Each prefix mass is obtained in at most two additional cut queries using

$$2\mu_A^{\mathrm{out}}(X) = d_A + c_G(X \setminus A) - c_G(A \cup X),$$
$$2\mu_A^{\mathrm{in}}(X) = d_A + c_G(A \cap X) - c_G(A \setminus X).$$

Lemma 23 (Completion without a depth factor)**.** *For $0 < \eta < 1$, $O(n\eta^{-1} \log n)$ deterministic cut queries construct a nonnegative weighted graph F with $c_F(A) = d_A$ for every cluster and*

$$|M_F(A, B) - M_G(A, B)| \le \eta \min(d_A, d_B), \qquad M_G(A, B) = \mathbf{1}_A^{\mathsf{T}} L_G \mathbf{1}_B. \tag{29}$$

The graph F can be chosen with $O(n\eta^{-1})$ positive-weight edges.

Proof. Apply Lemma 22 to both measures of every cluster, with $q = \lceil 2/\eta \rceil$. Find F by nonnegative linear feasibility, requiring all cluster cuts to equal d_A and all the recorded boundary prefix masses to have their recorded values. These are linear equality constraints in the weights of F, and G is a feasible solution.

For fixed A, every interaction $M_G(A, B)$ is one of

$$-\mu_A^{\mathrm{out}}(B), \qquad \mu_A^{\mathrm{in}}(B), \qquad d_A - \mu_A^{\mathrm{out}}(B),$$

according as B is disjoint from, contained in, or contains A; the diagonal is already exact. The same identities hold for F. Since B is an interval, Lemma 22 bounds the error by ηd_A. Swapping A, B bounds it by ηd_B, giving (29).

There are $O(nq)$ equality constraints. The feasible polytope is bounded because it fixes every singleton weighted degree. An extreme point has at most the rank of the constraint matrix many positive coordinates: otherwise there is a nonzero dependence supported on its positive coordinates, and sufficiently small perturbations in both directions preserve feasibility. Choose an extreme point deterministically to obtain $O(nq)$ positive edge weights. □

9.2 Representation cost controls completion error

Lemma 24 (Error and domination)**.** *For every nontrivial cut S,*

$$|c_F(S) - c_G(S)| \le \eta \frac{R(S)^2}{\lambda}. \tag{30}$$

Moreover $R(S) \ge c_G(S)$.

Proof. Take a minimizing binary labeling in (27). Let $A_1, \ldots, A_k$ be the clusters whose parent edge has different endpoint labels, and let $\sigma_i \in \{-1, 1\}$ be the change of label across that edge, from parent to child. Telescoping along each root path gives

$$\mathbf{1}_S = \ell(\mathrm{root})\mathbf{1}_V + \sum_{i=1}^{k} \sigma_i \mathbf{1}_{A_i}, \qquad R(S) = \sum_{i=1}^{k} d_{A_i}. \tag{31}$$

A cut edge whose cluster is $\emptyset$ or V has $d_{A_i} = 0$ and $\mathbf{1}_{A_i} \in \{0, \mathbf{1}_V\}$; absorb its term into the constant multiple of $\mathbf{1}_V$ and drop it from the list, which changes neither $R(S)$ nor the sums below. Every remaining A_i is a nontrivial shore of G, so $d_{A_i} \geq \lambda$ and $k \leq R(S)/\lambda$. The Laplacians annihilate $\mathbf{1}_V$, so (29) gives

$$\begin{aligned} |c_F(S) - c_G(S)| &\leq \eta \sum_{i,j=1}^{k} \min(d_{A_i}, d_{A_j}) \\ &\leq \eta k \sum_{i=1}^{k} d_{A_i} \leq \eta R(S)^2/\lambda. \end{aligned}$$

Finally, every edge of G crossing S must cross at least one of the sets A_i, by (31). Summing nonnegative edge weights gives $c_G(S) \leq \sum_i c_G(A_i) = R(S)$. □

9.3 An explicit graph preventing artificial small cuts

Lemma 25 (Terminal retraction of a tree)**.** *Let T be a finite tree with a nonempty terminal set, nonnegative edge capacities c_e, and nonnegative edge lengths ℓ_e. There is a map f from its vertices to its terminals, fixing every terminal, such that*

$$\sum_{xy \in E(T)} c_{xy} d_\ell(f(x), f(y)) \leq 42 \sum_{e \in E(T)} c_e \ell_e. \tag{32}$$

Proof. First assume all lengths are positive. Root the tree, let $h(x)$ be the length of the root path, and let $r(x)$ be the distance from x to its nearest terminal. Fix an order for breaking nearest terminal ties. Choose independent uniform shifts $s, t \in [0, 1)$. For a nonterminal x set

$$j(x) = \lfloor \log_2 r(x) + s \rfloor, \qquad \Delta(x) = 2^{j(x)-s}, \qquad b_\Delta(x) = \lfloor 2h(x)/\Delta + t \rfloor.$$

Thus $\Delta(x) \leq r(x) < 2\Delta(x)$. At this fixed scale, climb from x to the highest ancestor in its same height band, and map x to a nearest terminal of that ancestor. Map every terminal to itself. The climb has length less than $\Delta(x)/2$; the triangle inequality and the fact that r is 1-Lipschitz give

$$d_\ell(x, f(x)) \leq 2r(x).$$

Consider a tree edge xy of length δ, and put $r_0 = \min\{r(x), r(y)\}$. Every choice of shifts satisfies

$$d_\ell(f(x), f(y)) \leq 2r(x) + \delta + 2r(y) \leq 4r_0 + 3\delta.$$

If $\delta \geq r_0$ this is at most 7δ. Otherwise both endpoints are nonterminals. Their scale indices differ with probability at most

$$|\log_2 r(x) - \log_2 r(y)| \leq 2\delta/r_0.$$

Conditional on equal scale indices, their common scale satisfies $\Delta > r_0/2$. The independent band shift separates their bands with probability at most $2\delta/\Delta \leq 4\delta/r_0$. Here we used the elementary fact that a uniformly shifted unit grid separates two real numbers with probability at most their distance. If neither event occurs, the endpoints have the same highest ancestor in their band and hence the same image under f. The union bound and $4r_0 + 3\delta < 7r_0$ therefore give

$$\mathbb{E}\, d_\ell(f(x), f(y)) \leq 42\delta.$$

Multiplying by capacities and summing proves that some map satisfies (32). Only finitely many maps are possible.

For nonnegative lengths replace ℓ by $\ell+\varepsilon$. The minimum, over the same finite set of terminal-fixing maps, of the left side measured in the original metric is at most $42\sum_e c_e\ell_e + 42\varepsilon\sum_e c_e$ for every $\varepsilon>0$. Letting ε tend to zero proves (32). Zero capacities require no division and cause no exception. □

Lemma 26 (A graph approximating the terminal cuts of a tree)**.** *For the known capacitated hierarchy tree there is an explicit nonnegative weighted graph J_{prot} on the terminal set V such that*

$$R(S) \le c_{J_{\mathrm{prot}}}(S) \le 42R(S) \qquad (S \subseteq V). \tag{33}$$

The graph can be found deterministically without queries to G. More generally we write $\rho \ge 1$ for a valid guard quality; we may always take the absolute constant $\rho_0 = 42$.

Proof. For each terminal-fixing map f, replace a tree edge xy by the terminal edge $f(x)f(y)$ with the same capacity, dropping loops. Call this terminal graph J_f, and route its edges on their unique tree paths. Let $\mathrm{load}_f(e)$ be the load on tree edge e. Exchanging two finite sums gives

$$\sum_e \ell_e \,\mathrm{load}_f(e) = \sum_{xy\in E(T)} c_{xy} d_\ell(f(x),f(y)).$$

Apply finite minimax to the matrix with rows f, columns e, and entries $\mathrm{load}_f(e) - 42c_e$. For every probability distribution on columns, Lemma 25, with those probabilities as lengths, supplies a row of nonpositive average. Consequently there is a distribution p on the finite set of maps with $\sum_f p_f \,\mathrm{load}_f(e) \le 42c_e$ for every edge. This statement includes edges of capacity zero.

Put $J_{\mathrm{prot}} = \sum_f p_f J_f$. Pulling a terminal cut back through f gives a feasible tree labeling of cost $c_{J_f}(S)$, so averaging gives $R(S) \le c_{J_{\mathrm{prot}}}(S)$. Conversely fix a minimum tree extension of S. Every routed terminal edge crossing S uses at least one tree edge cut by this extension. Its expected load bounds give $c_{J_{\mathrm{prot}}}(S) \le 42R(S)$. A tree with no edges is immediate.

For deterministic computation, enumerate the terminal-fixing maps and solve the finite feasibility linear program for p just described, or enumerate terminal cuts and solve directly for edge weights satisfying (33). All its data are known. The random shifts prove existence only; the query algorithm uses deterministic finite optimization. The construction and its analysis follow the framework of Englert et al. [EGK+14, Sections 3–4], who combine terminal-fixing maps with the same transfer from distance bounds to congestion bounds; their Theorem 3.7 gives constant-quality flow sparsifiers of weighted trees, and hence (33) with some absolute constant. We include the argument to obtain an explicit constant and a cut-only statement. □

Proof of Theorem 21. Query the cluster capacities and compute J_{prot}, ρ from Lemma 26. Put

$$\eta = \frac{1}{100\rho^2 B^2}, \qquad H_0 = F + \sqrt{\eta}\, J_{\mathrm{prot}},$$

where F is the completion from Lemma 23. For a nontrivial cut put $x = \sqrt{\eta}\, R(S)/\lambda(G)$. The completion error, nonnegativity of F, and the lower bound in (33) imply

$$\frac{c_{H_0}(S)}{\lambda(G)} \ge \max\{0, 1-x^2\} + x \ge 1.$$

For every minimum cut S^*, which satisfies the promise, the upper bounds give

$$\frac{c_{H_0}(S^*)}{\lambda(G)} \le 1 + \eta B^2 + \sqrt{\eta}\,\rho B = 1 + \frac{1}{100\rho^2} + \frac{1}{10} \le \frac{111}{100}.$$

Thus H_0 satisfies (25) and is a weak skeleton. Since $\rho = O(1)$, the number of queries is $O(n\eta^{-1}\log n) = O(nB^2\log n)$. □

Remark 27 (An elementary alternative). For a hierarchy of height h, one can replace Lemma 26 by eliminating the internal vertices of its Laplacian. The resulting Schur-complement graph J_0 on the leaves satisfies

$$R(S)/(2h) \le c_{J_0}(S) \le R(S).$$

Indeed, its cut value is the minimum energy of real internal potentials extending the leaf labels. A binary minimizing labeling gives the upper bound. For the lower bound, decompose a maximum flow of value $R(S)$ between the terminal shores into paths. Each path has length at most $2h$ and endpoint potential difference one, so its sum of squared potential differences is at least $1/(2h)$ by Cauchy–Schwarz. Summing over the capacity-respecting flow paths bounds the total energy below by $R(S)/(2h)$. Thus $J_{\mathrm{prot}} = 2hJ_0$ satisfies $R(S) \le c_{J_{\mathrm{prot}}}(S) \le 2hR(S)$, giving guard quality $\rho = 2h$. The same proof gives $O(nh^2B^2\log n)$ queries without a terminal sparsification theorem.

10 Coarse completion and hierarchy refinement

We improve the quadratic dependence on the supplied hierarchy bound in Theorem 21. The refinement uses a tree approximation of an explicit graph. We state that offline ingredient first, use it to prove the refinement bound, and discuss it in Section 10.4.

10.1 Coarse completion of the current hierarchy

The one-step construction in Theorem 21 uses quadratically many queries in B. A coarser completion can instead be used to improve the hierarchy. The important choice is to give the guard more weight than the minimal value $\sqrt{\eta}$ used for the final skeleton.

The offline ingredient is the following lemma, due to Räcke and Shah [RS14]; see Section 10.4. Put

$$L(r) = \lceil \log_2(2r) \rceil \quad (r \ge 1), \qquad L(0) = 1, \qquad K(r) = L(L(r)).$$

Thus $K(r) = \lceil \log_2(2L(r)) \rceil$, and $L(r)K(r) \ge 1$ always. In natural logarithms, $L(r) \le 3\log(2r)$ and $K(r) \le 7\log\log(2r)$ for $r \ge 2$, so $K(r) = O(\log\log(2r))$.

Lemma 28 (An offline hierarchy). *There is an absolute constant a_0 such that for every explicit nonnegative weighted graph J on $r \ge 1$ vertices there is a canonical hierarchy $\mathcal{T}_J$ with at most $2r+1$ nodes, height at most $2L(r)$, and*

$$c_J(S) \le R_{\mathcal{T}_J}(S) \le a(r)c_J(S) \quad (S \subseteq V(J)), \qquad a(r) = a_0L(r)K(r). \tag{34}$$

It can be chosen deterministically from explicit J without queries to any other graph. No connectivity or positive-weight assumption is required.

Fix a known uniform bound $a = a(n) \ge 1$ in Lemma 28, with $a(n) = a_0L(n)K(n)$. Fix also the absolute constant $\rho_0 = 42$ such that the guard of Lemma 26 can always be chosen to satisfy

$$R(S) \le c_{J_{\mathrm{prot}}}(S) \le \rho_0 R(S) \qquad (S \subseteq V). \tag{35}$$

These bounds concern explicit graphs and trees. They are independent of the weights and of the input minimum-cut value.

10.2 Transferring a new hierarchy back to the input

Suppose the current canonical hierarchy has terminal function R and satisfies $R(S^*) \le B\lambda$ for every minimum cut S^*; fix one such S^* for the analysis. Obtain a nonnegative completion F by Lemma 23 with parameter η and put

$$H = F + \gamma J_{\mathrm{prot}}, \qquad \gamma > 0.$$

The graph H is explicit and has connected positive support, since J_{prot} dominates R, and $R(S) \ge c_G(S) > 0$ for every nontrivial cut. An offline hierarchy of H has canonical capacities $c_H(A)$ and terminal function R_H satisfying $R_H(S) \le a c_H(S)$. Keep its topology, query $c_G(A)$ for each of its nonroot clusters, and denote the resulting canonical terminal function by R_{new}.

Lemma 29 (Hierarchy transfer)**.** *For every nontrivial cut S,*

$$c_G(S) \le c_H(S) + \frac{\eta}{\gamma^2\lambda} c_H(S)^2, \tag{36}$$

$$R_{\mathrm{new}}(S) \le R_H(S) + \frac{\eta}{\gamma^2\lambda} R_H(S)^2. \tag{37}$$

In particular, if

$$x = a(1 + \eta B^2 + \gamma\rho_0 B), \tag{38}$$

then

$$R_{\mathrm{new}}(S^*) \le \left(x + \frac{\eta}{\gamma^2} x^2\right)\lambda. \tag{39}$$

The new hierarchy requires at most $2n$ additional cut queries.

Proof. Nonnegativity of F and (35) give $c_H(S) \ge \gamma R(S)$. Lemma 24 therefore implies

$$c_G(S) \le c_F(S) + \eta R(S)^2/\lambda \le c_H(S) + \eta c_H(S)^2/(\gamma^2\lambda),$$

proving (36).

Take an optimal labeling of the H-hierarchy for S. Let $A_1, \dots, A_m$ be the clusters whose parent edges it cuts and write $h_i = c_H(A_i) \ge 0$. Use this same labeling after querying the G-capacities. Applying the preceding inequality to each A_i gives

$$\begin{aligned} R_{\mathrm{new}}(S) &\le \sum_{i=1}^{m} c_G(A_i) \\ &\le \sum_{i=1}^{m} h_i + \frac{\eta}{\gamma^2\lambda} \sum_{i=1}^{m} h_i^2 \\ &\le R_H(S) + \frac{\eta}{\gamma^2\lambda} R_H(S)^2. \end{aligned}$$

This is a bound on the sum of the queried capacities; no term is multiplied by the number of clusters or the hierarchy depth.

For the fixed minimum cut S^*, the old completion bound gives

$$c_H(S^*) \le (1 + \eta B^2 + \gamma\rho_0 B)\lambda, \qquad R_H(S^*) \le x\lambda.$$

Substitute this in (37). There are at most $2n$ nonroot clusters, proving the query count. □

10.3 Halving distortion with linearly many queries in its bound

Lemma 30 (One refinement round)**.** *Set*

$$\gamma = \frac{1}{100a^2\rho_0^2}. \tag{40}$$

If $B \geq 1/\gamma$, a canonical hierarchy with promise $B\lambda$ for every minimum cut can be replaced by one with promise $(B/2)\lambda$ for every minimum cut, using $O(na^2B\log(2n))$ cut queries.

Proof. Use the preceding construction with $\eta = \gamma/B$. In particular $0 < \eta < 1$. Since $\gamma B \geq 1$ and $\rho_0 \geq 1$, the value in (38) satisfies

$$x = a(1 + \gamma B + \gamma\rho_0 B) \leq 3a\rho_0\gamma B.$$

Consequently

$$\begin{aligned} x + \frac{\eta}{\gamma^2}x^2 &= x + \frac{x^2}{\gamma B} \\ &\leq (3a\rho_0 + 9a^2\rho_0^2)\gamma B \\ &\leq 12a^2\rho_0^2\gamma B = \frac{12}{100}B < \frac{B}{2}. \end{aligned}$$

Lemma 29 proves the new promise. The completion costs $O(n\eta^{-1}\log(2n)) = O(na^2B\log(2n))$ queries, since ρ_0 is an absolute constant. The guard and the H-hierarchy use explicit data only, and querying the new cluster capacities adds $O(n)$ queries. Neither S^* nor λ is queried or used to choose the parameters. □

Theorem 31 (Linear dependence on the hierarchy bound)**.** *Suppose $\lambda > 0$ and a canonical hierarchy on $n \geq 2$ vertices is supplied, together with $B \geq 1$ such that $R(S^*) \leq B\lambda$ for every minimum cut S^*. A weak skeleton can be constructed deterministically using*

$$O\Big(nB\log^3(2n)(\log\log(2n))^2\Big)$$

cut queries, independently of hierarchy depth and weight spread. Offline computation may be exponential.

Proof. Query the initial cluster capacities if they are not already known. Keep a, ρ_0, γ fixed. While the current promised bound is at least $1/\gamma$, apply Lemma 30, and replace B by $B/2$. Every original minimum cut satisfies every new promise, so the process is valid without ever identifying these cuts. The sum of the successive promised bounds is at most twice the initial one. Refinement therefore costs $O(na^2B\log(2n))$ queries in total.

At termination let $b < 1/\gamma$ be the remaining bound. Apply Theorem 21 to this last hierarchy. Its cost is

$$O(nb^2\log(2n)) = O(nb\gamma^{-1}\log(2n)) = O(na^2B\log(2n)).$$

This also covers the case where the initial B is below the stopping threshold and no refinement is needed. Finally $a^2 = a_0^2L(n)^2K(n)^2 = O(\log^2(2n)(\log\log(2n))^2)$, giving the stated bound. The final application of Theorem 21 supplies all the weak-skeleton guarantees; intermediate completions need not themselves be weak skeletons.

The query parameters can be integers throughout: use $q = \lceil 2B/\gamma \rceil$ in a refinement round and $q = \lceil 200\rho_0^2b^2 \rceil$ in the final completion. Then the completion error parameter $2/q$ is no larger than the one used in the corresponding proof. The ceilings change only absolute constants. Because the

promised bounds sum to at most $2B$, no factor $\log B$ is charged for the number of rounds. With the fixed value $\gamma^{-1} = 100a_0^2\rho_0^2 L(n)^2 K(n)^2$, this is a public $O(nBL(n)^3K(n)^2)$ budget determined by n, B and the supplied hierarchy, before the hidden graph is known. □

Remark 32 (All minimum cuts at once). The universal promise costs nothing. The construction never reads a minimum cut: every parameter depends on B and explicit data, and each refinement round and the final completion are fixed before the analysis picks an S^*. The proofs of Theorem 21, Lemma 30 and Theorem 31 are therefore pointwise in S^*, and a single run yields (25) for every minimum cut simultaneously. The tree cut sparsifier of Theorem 7 satisfies $R(S) \le \alpha\, c(S)$ for every shore, so the promise holds for every minimum cut of the quotient.

Theorem 6 follows. Bring the given B-tree cut sparsifier into normal form as in Section 4; this is a canonical hierarchy with $O(n)$ nodes and $c_G(S) \le R(S) \le Bc_G(S)$ for every shore, so the promise $R(S^*) \le B\lambda$ holds for every minimum cut, and Theorem 31 gives the weak skeleton.

Remark 33 (Calibrating the constants from explicit data). The constants a_0 and $\rho_0 = 42$ above suffice. Alternatively, an implementation can calibrate smaller constants from explicit data. Use the guard's computed quality ρ, maintain a trial hierarchy distortion $\widehat{a} \ge 1$, initially 1, and set $\gamma = (100\widehat{a}^2\rho^2)^{-1}$. If $B < 1/\gamma$, finish with Theorem 21. Otherwise construct F, H as above and measure the chosen H-hierarchy's exact distortion offline. If it exceeds $\widehat{a}$, double $\widehat{a}$ and retry; if it does not, query the new cluster capacities and halve B. Choose the H-hierarchy by the minimization in Lemma 28, so the trial bound never exceeds $2\max\{1, a(n)\}$. Also $\rho \le \rho_0$. The retry costs are geometric in $\widehat{a}^2$, while accepted rounds are geometric in B, giving the same total query bound. All tests in this implementation use known explicit graphs.

10.4 Offline hierarchies of explicit graphs

Räcke and Shah [RS14] show that every graph has a tree cut sparsifier of quality $O(\beta \log n \log\log n)$, where β is the approximation factor of the sparsest-cut routine used in their construction. With exact sparsest cuts, $\beta = 1$, and this is the existential bound of Lemma 28; their polynomial-time construction uses $\beta = O(\sqrt{\log n})$. Their tree is a hierarchy with singleton leaves in which the children of each cluster partition it into parts of at most half its size, so it has at most $2r - 1$ nodes and height less than $L(r)$, and the normalization of Section 4 gives it canonical capacities without increasing its distortion. A formalized, cut-only version of their argument, with exact finite minimization in place of sparsest-cut approximation, gives $a_0 = 268$; it is stated with the slack of Lemma 28, at most $2r + 1$ nodes and height at most $2L(r)$, and its nodes of $V(J)$ are childless but need not be the only leaves. Since J is explicit, the lemma can be applied deterministically: enumerate all hierarchies with these labeled leaves and no unary nodes, and choose one minimizing its worst cut distortion, with fixed tie-breaking. A positive tree cost on a zero J-cut has infinite distortion; a zero cost on both sides imposes no constraint. The lemma guarantees a finite answer obeying (34), even for disconnected positive support or the all-zero graph.

11 Discovering a hierarchy for arbitrary weights

We now construct an initial hierarchy to which Theorem 31 applies. Weighted degree normalization gives a polynomial ratio between the largest weighted degree and a witnessed cut bound. Residual rounding then either verifies the required hierarchy bound or supplies a smaller witnessed cut bound that forces the next normalization to halve the graph. This test is computed from explicit data and queried cut values; it is not the minimum-cut certificate of Section 7.

11.1 Terminal compression of explicit trees

Lemma 34 (Terminal compression of a tree)**.** *Let T be a finite tree with nonnegative edge capacities and let C be a nonempty set of its vertices, with $q = |C|$. There is a fixed contraction map $\pi : V(T) \to V(T')$, injective on C, with*

$$|V(T')| \le \max\{1, 2q-2\},$$

such that, for every assignment of binary labels to C, some minimum-cost extension to T is constant on each fiber of π. The contraction is computable from the explicit tree and is independent of the terminal labels.

Proof. Repeatedly contract leaves outside C into their neighbors. This prunes the tree down to the minimal subtree joining C; every removed branch can be labeled uniformly with its attachment vertex at zero cost. If $q = 1$, this leaves one vertex and proves the assertion.

For $q \ge 2$, every leaf of the pruned tree lies in C. On each maximal path whose internal vertices lie outside C and have degree two, choose a minimum-capacity edge, breaking ties deterministically, and contract all other edges of the path. For fixed endpoint labels, the optimum path cost is zero if the labels agree and the minimum edge capacity if they differ. Both optima are attained with all contracted edges uncut. The choices on different paths are compatible, so the fixed contraction preserves the minimum extension cost for every labeling of C. In the resulting tree T' with n' vertices, every vertex outside C has degree at least three and every vertex of C has degree at least one. Summing degrees gives $2(n'-1) \ge q + 3(n'-q)$, so $n' \le 2q-2$. The argument also applies when some capacities are zero, and the terminals need not be leaves. □

11.2 Safe contractions and weighted degree normalization

We work on a current quotient of the input. Write N for its number of vertices, G for this quotient, and $\delta = \min_v d_v$. Every query to G is one original query on a union of contracted classes. We maintain all weighted degrees and a known nontrivial cut of value Λ, with

$$0 < \lambda \le \Lambda \le \delta. \tag{41}$$

Initially query the weighted degrees and use a singleton of minimum weighted degree. An individually learned edge with weight strictly greater than Λ can be contracted: no cut of value at most Λ, including every minimum cut and the witness for Λ, can separate its endpoints.

Lemma 35 (Weighted degree normalization)**.** *Using safe contractions, one can obtain a quotient with N' vertices and an updated witnessed bound $\Lambda' \le \Lambda$ such that*

$$0 < \lambda \le \Lambda' \le \min_v d_v, \qquad \max_v d_v \le (N'-1)\Lambda'. \tag{42}$$

Every original minimum cut survives. With the weighted degrees already known, the query cost is $O((N-N')\log(2N))$. Across all invocations in one algorithm, the total normalization cost is $O(n \log(2n))$.

Proof. While some vertex v has $d_v > (N-1)\Lambda$, find an edge vu of weight greater than Λ as follows. Start with candidates $W = V \setminus \{v\}$ and known total mass $w_G(v, W) = d_v$. Split W into two nearly equal nonempty parts. Evaluate one part's mass using

$$2w_G(v, X) = d_v + c_G(X) - c_G(X \cup \{v\}), \qquad v \notin X,$$

and obtain the other's by subtraction. Retain the part with larger average mass per candidate, with fixed tie-breaking. Its average is at least the old average, which is strictly greater than Λ. After $O(\log(2N))$ queries, the remaining singleton gives $w_G(v,u) > \Lambda$.

Contract u, v. No cut of value at most Λ separates them, so all minimum cuts and the current witness survive. The new vertex has weighted degree

$$d_{uv} = d_u + d_v - 2w_G(u,v). \tag{43}$$

Every other vertex keeps exactly the same weighted degree. Thus no additional queries are needed to update the entire weighted degree list. Replace Λ by $\min\{\Lambda, d_{uv}\}$, using the new singleton shore as witness if its weighted degree is smaller, and decrement N. The new witnessed bound still satisfies $\lambda \le \Lambda \le \min_v d_v$.

Repeat with the new values of N and Λ. The witness remains nontrivial, so at least two vertices remain. The stopping condition is exactly (42). There is one search per contraction, proving the query bound. Over the entire algorithm there are at most $n-2$ contractions, even when the witnessed bound changes many times. □

The lemma does not assert that every remaining edge has weight at most Λ'. For example, on vertices $0, 1, 2, 3$ give edges $01, 02, 12, 23$ weights $2, 1, 1, 1$, respectively. The minimum weighted degree and the witnessed bound can both be 1, and every weighted degree is at most 3, so normalization does nothing although edge 01 has weight 2. Such edges may be retained in the explicit graph K during the next phase.

Lemma 36 (A large bound decrease forces halving)**.** *Suppose the current graph has $N \ge 2$ vertices and all weighted degrees are at least a witnessed bound Λ. If a new nontrivial cut gives a bound $u < \Lambda/(4N)$, as in Lemma 40, then normalization started with u leaves at most $\lfloor N/2 \rfloor$ vertices.*

Proof. Throughout this normalization, the current bound Λ' is at most u and the number of vertices is at most N. Every witnessed bound is the value of a nontrivial cut, so $\Lambda' \ge \lambda > 0$. Consider an old vertex that never participates in a contraction. Its weighted degree is unchanged and at least Λ. At termination the weighted degree condition would require

$$\Lambda \le d_v \le (N' - 1)\Lambda' \le (N-1)u < \Lambda,$$

a contradiction; the third inequality uses $\Lambda' \ge 0$. Every final vertex therefore contains at least two vertices from the graph at the start of this invocation. These classes partition the old vertices, proving the claim. Further decreases of the bound during normalization only strengthen it. □

11.3 Learning a residual graph down to a weighted degree floor

Start each phase after weighted degree normalization, so

$$\Lambda \le d_v \le (N-1)\Lambda \qquad (v \in V(G)). \tag{44}$$

Fix an integer $2 \le k \le N-1$ and set

$$t = \frac{\Lambda}{16N^2}, \qquad L = 16\lceil N/k \rceil \lceil \log_2(2N) \rceil. \tag{45}$$

The parameter t is a residual weighted degree floor, and L bounds the number of centers. Both remain fixed during the phase. Contractions occur only during normalization between phases.

Lemma 37 (Residual preparation)**.** *Using $O(Nk\log^2(2N))$ cut queries, one can learn an exact subgraph $K \subseteq G$ such that, for $R = G - K$ and its known weighted degrees r_v,*

$$r_v \le t \quad \text{or} \quad w_R(v,u) \le r_v/k \quad \text{for every } u. \tag{46}$$

The graph K may contain edges heavier than Λ.

Proof. Start with K empty. Cut queries to R are simulated by subtracting the known contribution of K from a query to G. Its weighted degrees are updated without further queries whenever an edge is added to K.

A scan of a vertex v with $r_v > t$ sets $q_v = r_v$ and, keeping K fixed during the search, finds all residual incident edges heavier than $q_v/(2k)$. By Lemma 8, there are fewer than $2k$ such edges and the scan costs $O(k\log(2N))$ queries. Add their full weights to K and update the residual weighted degrees. Thus each edge of R is either an entire original edge or zero.

After a scan every remaining incident edge has weight at most $q_v/(2k)$, and subsequent removals preserve this upper bound. The relative lightness condition holds while $r_v \ge q_v/2$. Repeatedly scan any vertex with $r_v > t$ that has not been scanned or whose weighted degree has fallen below half its last reference value. At termination this proves (46).

Every first reference is at most $(N-1)\Lambda$. Successive references for a vertex decrease by a factor greater than two and stay above t. Since $(N-1)\Lambda/t = 16N^2(N-1)$, there are at most $1 + \log_2(16N^2(N-1)) = O(\log(2N))$ scans per vertex. This proves both finite termination and the query bound, independently of the weight spread. The argument uses only $\Lambda > 0$ and the upper bound $d_v \le (N-1)\Lambda$ in (44), not the lower bound $\Lambda \le d_v$; the count of edges found by one scan is likewise not needed for the query bound. □

11.4 Selecting centers with weighted degree buckets

Freeze K, R, and the residual weighted degrees after preparation. Learning center rows does not remove further edges from R. Let H_C^R be the subgraph of R containing exactly the edges incident to the selected center set C.

For possibly overlapping vertex sets A, W, write

$$m_R(A,W) = \sum_{v\in A}\sum_{u\in W} w_R(v,u).$$

This counts an edge inside $A \cap W$ twice. With $I = A \cap W$, $X = A \setminus W$, $Y = W \setminus A$, and $Z = V \setminus (A \cup W)$,

$$m_R(A,W) = \sum_{v\in I} r_v - w_R(I,Z) + w_R(X,Y). \tag{47}$$

The weighted degrees are known and the two disjoint-set masses cost at most six cut queries in total by (17).

Lemma 38 (Bucketed center selection)**.** *Let $2 \le k \le N-1$ be as in (45), so in particular $k \le 8N$. Using $O((N^2/k)\log(2N))$ further queries, one can learn all edges incident to a set C of at most L centers such that*

$$v \notin C \quad \Longrightarrow \quad r_v \le t \ \text{ or } \ w_R(v,C) \ge r_v/k. \tag{48}$$

Proof. Call a vertex active when

$$v \notin C, \qquad r_v > t, \qquad b_v := w_R(v,C) < r_v/k.$$

For an active vertex put $a_{vu} = kw_R(v,u)/(2r_v)$ and $p_v = \prod_{c\in C}(1-a_{vc})$. Relative lightness gives $0 \le a_{vu} \le 1/2$ and $p_v \ge 1 - \sum_{c\in C} a_{vc} > 1/2$. With $\Phi = \sum_{v \text{ active}} p_v$ and $\sigma(u) = \sum_{v \text{ active}} p_v a_{vu}$, we have

$$\sum_{u\notin C} \sigma(u) = \frac{k}{2} \sum_{v \text{ active}} p_v(1 - b_v/r_v) \ge \frac{k}{4}\Phi. \tag{49}$$

To evaluate group scores cheaply, use $t < r_v \le (N-1)\Lambda$ and $1/2 < p_v \le 1$ to bound $\theta_v = p_v/r_v$ by

$$\frac{1}{2(N-1)\Lambda} < \theta_v < \frac{1}{t}, \qquad \frac{2(N-1)\Lambda}{t} < 32N^3.$$

Round each coefficient down to a power of two times $[2(N-1)\Lambda]^{-1}$, so $\widetilde{\theta}_v \le \theta_v < 2\widetilde{\theta}_v$. There are $O(\log(2N))$ buckets A_j with common coefficient θ_j. The surrogate scores

$$\widetilde{\sigma}(u) = \frac{k}{2} \sum_{v \text{ active}} \widetilde{\theta}_v w_R(v,u)$$

satisfy $\sigma(u)/2 \le \widetilde{\sigma}(u) \le \sigma(u)$. For a candidate set W, their total is

$$\sum_{u\in W} \widetilde{\sigma}(u) = \frac{k}{2} \sum_j \theta_j m_R(A_j, W), \tag{50}$$

which costs $O(\log(2N))$ queries by (47). The total over $V \setminus C$ is also known directly from the r_v and b_v.

Keep the active set and the buckets fixed during a balanced search. Split the candidate set into two nearly equal parts, evaluate one part's total score, and obtain the other's by subtraction. Retain the part with larger average surrogate score, with fixed tie-breaking. The average never decreases. Since the surrogate scores are nonnegative, the initial average over the at most N candidates is at least $\frac{1}{N}\sum_{x\notin C} \widetilde{\sigma}(x)$, so the selected vertex u satisfies

$$\sigma(u) \ge \widetilde{\sigma}(u) \ge \frac{1}{N} \sum_{x\notin C} \widetilde{\sigma}(x) \ge \frac{k}{8N}\Phi.$$

This search costs $O(\log^2(2N))$ queries. Add u to C and learn its full original row with $O(N)$ queries, subtracting K to obtain its residual row.

A previously active vertex that stays active has its potential multiplied by $1 - a_{vu}$, so it decreases by $p_v a_{vu}$. Other previously active vertices leave the sum; since $0 \le a_{vu} \le 1/2$, their potentials p_v are nonnegative and at least $p_v a_{vu}$. No new vertex becomes active because C and each b_v only increase. Thus Φ decreases by at least $\sigma(u) \ge k\Phi/(8N)$. Initially $\Phi \le N$, whereas any nonempty active set has $\Phi > 1/2$. At most $8(N/k)\ln(2N) + 1 \le L$ selections therefore suffice. The total query cost is

$$O\left(\frac{N^2}{k}\log(2N) + \frac{N}{k}\log^3(2N)\right) = O\left(\frac{N^2}{k}\log(2N)\right).$$

When the active set is empty, (48) holds. □

11.5 Keeping known edges and rounding only the residual

Suppose preparation and center selection both finish. The explicit graph

$$H = K + H_C^R \subseteq G$$

has disjoint contributions from K and H_C^R. Use Lemma 28 to find a canonical hierarchy T_H approximating all its cuts with distortion $a_1 = O(\log(2N)\log\log(2N))$, $a_1 \geq 1$. Choose an arbitrary vertex v_0 and apply Lemma 34 with terminal set

$$T_0 = C \cup \{v_0\}.$$

The extra vertex ensures that this set is nonempty; no additional row is learned for it. The nonempty fibers on original vertices give a fixed partition $P_1, \ldots, P_p$, where

$$p \leq N, \qquad p \leq \max\{1, 2|T_0| - 2\} = O\left(1 + \frac{N}{k}\log(2N)\right). \tag{51}$$

For every cut S, minimum-cost extension of its labels on T_0 gives a union $\widehat{S}$ of these classes with

$$\widehat{S} \cap T_0 = S \cap T_0, \qquad c_H(\widehat{S}) \leq a_1 c_H(S). \tag{52}$$

To verify this, fix only the labels on T_0. The original labeling of all leaves costs at most $a_1 c_H(S)$, terminal compression permits a minimizing extension constant on fibers, and the canonical tree dominates H.

Put $X = S \triangle \widehat{S}$. No vertex in X belongs to C. Charge the vertices with $r_v \leq t$ at most Nt in total, and use (48) on the others. Since $V \setminus C$ is independent in H_C^R,

$$\begin{aligned} \mathrm{vol}_R(X) &\leq k\,\mathrm{vol}_{H_C^R}(X) + Nt = k c_{H_C^R}(X) + Nt \\ &\leq k(c_{H_C^R}(S) + c_{H_C^R}(\widehat{S})) + Nt \\ &\leq k(1 + a_1) c_G(S) + Nt. \end{aligned} \tag{53}$$

There is no corresponding bound claimed for $\mathrm{vol}_G(X)$ or $\mathrm{vol}_K(X)$. The known edges will be retained explicitly instead.

Learn the quotient $Q = G/\{P_1, \ldots, P_p\}$ with $O(p^2)$ queries. This partition is used for rounding and defining the auxiliary graph. Safe contractions in the algorithm are performed only by Lemma 35 on the current graph G. Construct a graph J_{aux} with terminal set $V(G)$ and one additional vertex z_i for each part P_i, as follows:

1. Keep the edges of K between the original vertices.
2. Put the quotient graph Q on $z_1, \ldots, z_p$.
3. For $v \in P_i$, add an edge vz_i of weight r_v.

All its weights are known. For $S \subseteq V(G)$, let $\Psi(S)$ be its minimum terminal-extension cut value in J_{aux}. Equivalently,

$$\Psi(S) = c_K(S) + \min_{\widehat{S} \text{ a union of } P_i} \left(c_G(\widehat{S}) + \mathrm{vol}_R(S \triangle \widehat{S})\right). \tag{54}$$

Lemma 39 (Auxiliary graph with an additive error)**.** *Set $A = (1 + a_1)(2k + 1)$, so that $A = O(k\log(2N)\log\log(2N))$. For every cut S,*

$$c_G(S) \leq \Psi(S) \leq A c_G(S) + 2Nt. \tag{55}$$

The explicit graph J_{aux} has at most $2N$ vertices and requires only $O((N^2/k^2)\log^2(2N))$ queries after residual preparation and center selection.

Proof. For any union $\widehat{S}$ in (54), changing the labels in $X = S \triangle \widehat{S}$ changes an R-cut by at most $\mathrm{vol}_R(X)$. Thus

$$c_G(\widehat{S}) + \mathrm{vol}_R(X) \geq c_R(\widehat{S}) + \mathrm{vol}_R(X) \geq c_R(S).$$

Adding $c_K(S)$ and minimizing proves the lower bound.

For the upper bound, use the rounded cut from (52). We have

$$c_K(\widehat{S}) \leq a_1 c_G(S), \qquad c_R(\widehat{S}) \leq c_R(S) + \mathrm{vol}_R(X).$$

Substituting $c_G(\widehat{S}) = c_K(\widehat{S}) + c_R(\widehat{S})$ in (54) and using $c_K(S) + c_R(S) = c_G(S)$, then applying (53), gives

$$\Psi(S) \leq (1 + a_1) c_G(S) + 2\,\mathrm{vol}_R(X) \leq A c_G(S) + 2Nt.$$

The size and query claims follow from (51). All hierarchy choices and tree compression steps operate on already explicit graphs. □

11.6 A test that does not require knowing the optimum

Compute offline the minimum terminal-separating cut of J_{aux}:

$$\mu = \min_{\varnothing \neq S \subsetneq V(G)} \Psi(S), \qquad \Psi(S_{\mathrm{aux}}) = \mu. \tag{56}$$

The terminal side S_{aux} is nontrivial. The unrestricted global minimum cut of J_{aux} would not suffice here, since it might isolate only added vertices. Enumeration of terminal labelings and their extensions is one valid deterministic implementation. Lemma 39 implies

$$\lambda \leq \mu \leq A\lambda + 2Nt. \tag{57}$$

Lemma 40 (Certify a hierarchy or halve the graph)**.** *If $\mu \geq 4Nt$, then $O(N)$ additional queries construct a canonical hierarchy of G whose representation cost R_G satisfies*

$$c_G(S) \leq R_G(S) \leq B c_G(S) \quad \textit{for every } S \subseteq V(G), \qquad B = O(k \log^2(2N)(\log\log(2N))^2),$$

with an explicit valid value of B supplied to the algorithm. If $\mu < 4Nt$, one additional query gives a new witnessed bound

$$\Lambda' = c_G(S_{\mathrm{aux}}) \leq \mu < \frac{\Lambda}{4N}. \tag{58}$$

Weighted degree normalization with this new bound leaves at most $\lfloor N/2 \rfloor$ vertices before the next phase.

Proof. Suppose first that $\mu \geq 4Nt$. Combining this with (57) gives $2Nt \leq A\lambda$. Since $c_G(S) \geq \lambda$ for every nontrivial cut, the additive error is absorbed uniformly:

$$\Psi(S) \leq A c_G(S) + 2Nt \leq 2A c_G(S) \qquad (\varnothing \neq S \subsetneq V(G)). \tag{59}$$

For trivial cuts, $\Psi(S) = c_G(S) = 0$.

Apply Lemma 28 to the explicit graph J_{aux} on all its vertices, obtaining a tree of distortion $a_2 = O(\log(2N)\log\log(2N))$, $a_2 \geq 1$. Every vertex of J_{aux} is a leaf of this tree, since in a hierarchy the nodes of the vertex set have no children; in particular every terminal in $V(G)$ already has degree one. In this tree leave the leaves corresponding to the added vertices of J_{aux} free, and apply Lemma 34 with terminal set $V(G)$. The resulting tree T' has $O(N)$ vertices and preserves these

terminal-extension costs. Minimizing the original tree inequalities over the labels of the added leaves proves

$$c_G(S) \le \Psi(S) \le R_{T'}(S) \le a_2\Psi(S). \tag{60}$$

Compression contracts only leaves outside $V(G)$ and shortens terminal-free paths, so each terminal keeps degree one and every remaining leaf is a terminal: the leaves of T' are exactly $V(G)$. Root it at an internal vertex; for two terminals subdivide its single edge into two edges of the same capacity and use the new vertex as root.

For an edge e of this rooted tree, let A_e be its descendant terminal set. Query $c_G(A_e)$ and replace the edge capacity by this number. This never increases a capacity: cutting e alone is a feasible extension of the terminal cut A_e, so

$$c_G(A_e) \le R_{T'}(A_e) \le \mathrm{cap}_{T'}(e).$$

The resulting canonical hierarchy dominates every G-cut by the usual cut triangle inequality, and its extension costs are at most those of T'. By (59) and (60), the explicit choice $B = 2a_2A$ works. The values a_1, a_2 can be computed by exhaustive evaluation of the distortions of the chosen explicit hierarchies. There are only $O(N)$ newly queried cluster capacities.

Now suppose $\mu < 4Nt$. The lower inequality in (55) gives $c_G(S_{\mathrm{aux}}) \le \mu$. Since $4Nt = \Lambda/(4N)$, querying this nontrivial cut proves (58). Before the decrease every weighted degree is at least Λ, so Lemma 36 applies. The next phase therefore starts on at most half as many vertices, independently of the weight spread. □

11.7 The tree cut sparsifier tradeoff

Proof of Theorem 7. Fix an absolute constant $D \ge 1$ large enough that a successful phase on N vertices with parameter k produces a canonical hierarchy of distortion at most $DkL(N)^2K(N)^2$, by Lemma 40, and that the offline hierarchy of Lemma 28 has distortion at most $DL(N)K(N)$. Choose the constant C in the theorem large enough that its lower bound on α implies $\alpha \ge 2DL(n)^2K(n)^2$.

Query the weighted degrees, initialize Λ with a singleton of minimum weighted degree, and apply Lemma 35. On the current quotient with N vertices, set

$$k = \left\lfloor \frac{\alpha}{DL(N)^2K(N)^2} \right\rfloor \ge 2.$$

If $k \ge N$, reconstruct this quotient using $O(N^2)$ queries and compute its offline hierarchy. Its distortion is at most $DL(N)K(N) \le \alpha$, and the query cost is $O(N\alpha)$. Otherwise run a discovery phase with this k: residual preparation (Lemma 37), center selection (Lemma 38), reconstruction of the rounding quotient and assembly of the auxiliary graph (Lemma 39), and the test of Lemma 40. On success, its hierarchy has distortion at most $DkL(N)^2K(N)^2 \le \alpha$. On failure, Lemma 40 supplies a smaller witnessed bound; weighted degree normalization then at least halves N.

For the query bound, write $L = L(N)$ and $K = K(N)$. Adding the costs stated in the four lemmas, one phase through its test and canonicalization uses

$$O\left(NkL^2 + \frac{N^2}{k}L + \frac{N^2}{k^2}L^2 + N\right) \tag{61}$$

queries, the third term being the $O(p^2)$ reconstruction of the rounding quotient on p parts. By (51), the rounding partition has $p \le N$ parts and $p = O((N/k)L)$. Thus $p^2 \le Np = O(N^2L/k)$. Using

this bound for quotient reconstruction in (61), one phase costs

$$O\left(NkL^2 + \frac{N^2L}{k} + N\right) = O\left(N\alpha + \frac{N^2L^3K^2}{\alpha}\right),$$

since $\alpha/(2DL^2K^2) \le k \le \alpha/(DL^2K^2)$. The successive phase sizes decrease geometrically, so their sum is at most $2n$ and the sum of their squares is $O(n^2)$. Since $L(N) \le L(n)$ and $K(N) \le K(n)$, including a possible final reconstruction and the $O(n\log(2n))$ queries for all normalization calls gives

$$O\left(n\alpha + \frac{n^2L(n)^3K(n)^2}{\alpha} + nL(n)\right) = O\left(\frac{n^2}{\alpha}\log^3(2n)(\log\log(2n))^2\right)$$

queries in total, where the last step uses $\alpha^2 \le n$.

Let $\mathcal{P}$ be the final partition into contracted classes. Every contraction preserves every minimum cut, so $\mathcal{P}$ is safe. The final canonical hierarchy has at most $2N - 1 \le 8n$ vertices and is the required α-tree cut sparsifier of $G/\mathcal{P}$. The formalization omits the compression of Lemma 34 in Lemma 40 and keeps the hierarchy of J_{aux} restricted to the terminals $V(G)$, which has at most $2(N+p)+1 \le 8N$ nodes; this is the source of the bound $8n$. □

11.8 The complete algorithm and its query bound

Proof of Theorem 4. For $n = 2$, a singleton query returns a minimizing shore and its value, and the two singletons are the only shores. If $n < 2$, there is no proper shore; one may return no shore and value zero by convention, without queries. The theorem concerns $n \ge 2$.

First find a maximal forest of the positive support with $O(n\log(2n))$ queries by Lemma 9. If the support is disconnected, return value zero, a proper component as a minimizing shore, and the partition of V into components. A cut has value zero exactly when it is a union of components, so the proper unions are exactly the minimum-cut shores, including in the all-zero graph. Otherwise $\lambda = \lambda(G) > 0$.

Let C be the constant of Theorem 7, and fix an absolute integer $N_0 \ge 2$ such that

$$C\log^2(2n)(\log\log(2n))^2 \le \lfloor\sqrt{n}\rfloor \qquad (n \ge N_0).$$

Such an N_0 exists because every fixed power of a logarithm is $o(\sqrt{n})$, and it is fixed before any oracle answer is read. For $n < N_0$, reconstruct G from its singleton and pair cuts with $O(N_0^2) = O(1)$ queries and compute a minimum cut offline.

For $n \ge N_0$, apply Theorem 7 with $\alpha = \lfloor\sqrt{n}\rfloor$. Using $O((n^2/\alpha)\log^3(2n)(\log\log(2n))^2) = O(n^{3/2}\log^3(2n)(\log\log(2n))^2)$ queries, it returns a safe partition $\mathcal{P}$ of $V(G)$ and a canonical hierarchy of the quotient $G_q = G/\mathcal{P}$, on $N \le n$ vertices and with at most $8n$ nodes, with $c_{G_q}(S) \le R(S) \le \alpha c_{G_q}(S)$ for every quotient shore S (Section 11.7). Since $\mathcal{P}$ is safe, $N \ge 2$, $\lambda(G_q) = \lambda > 0$, and every minimum cut S_q^* of G_q satisfies $R(S_q^*) \le \alpha\lambda$. Now apply Theorem 31 to G_q with $B = \alpha$. Using $O(N\alpha\log^3(2N)(\log\log(2N))^2) = O(n^{3/2}\log^3(2n)(\log\log(2n))^2)$ further queries, each of them one query to G on a union of classes, it constructs a weak skeleton H_q of G_q: $\lambda(H_q) \ge \lambda$ and $c_{H_q}(S_q^*) \le (5/4)\lambda$ for every minimum cut S_q^* of G_q. The parameter balance happens here: learning costs $O((n^2/\alpha)\log^3(2n)(\log\log(2n))^2)$ and conversion $O(n\alpha\log^3(2n)(\log\log(2n))^2)$, and $\alpha = \Theta(\sqrt{n})$ equalizes them.

To produce a skeleton on the original n vertices, choose a fixed representative of each contracted class, realize the edges of H_q between these representatives, and connect the vertices inside each class by any explicit tree. Give every such internal tree edge capacity $2\lambda(H_q)$, a known positive

number. Every cut that splits a class has value at least $2\lambda(H_q)$; cuts respecting all classes have their exact H_q-values. Therefore the lifted graph H satisfies

$$\lambda(H) = \lambda(H_q) \geq \lambda.$$

Since $\mathcal{P}$ is safe, every original minimum cut is a union of classes and has its exact H_q-value, so every original minimum cut has H-value at most $(5/4)\lambda$. Thus H is a weak skeleton of G, obtained without further oracle queries.

Finally, Lemma 17 supplies $O(\log n)$ explicit trees covering every $(5/4)$-minimum cut of H, hence every minimum cut of G. The algorithm of Theorem 19 finds the minimum G-cut over these trees with $O(n \log^2 n)$ more queries. Query the returned shore once more, if its value has not already been recorded, to learn λ exactly; this adds at most one query. Then apply Theorem 52 with $\ell = \lambda$ to each tree and return λ together with the reported shores and their complements. Every minimum cut of G, or its complement, crosses at most two edges of some cover tree, so every minimum cut is reported, and every reported shore is a cut of G of value λ. A cut may be reported by several of the $O(\log n)$ trees, or by one tree as a shore and by another as its complement; the final list is deduplicated offline, at no query cost. This uses $O(n \log^2 n)$ further queries, independent of the number of minimum cuts.

All choices depend only on n, explicit graphs, and previous answers. The estimates give a fixed cap $C'n(\lfloor\sqrt{n}\rfloor + 1)L(n)^3K(n)^2$ for a universal constant C'. If desired, stop any inconsistent answer path at this cap and return failure. Correctness on every nonnegative weighted graph shows that an honest path never fails. Thus one deterministic decision tree is selected for each n before the input graph, and its cap holds for every answer function. □

Remark 41 (Where the logarithmic factors are saved). Suppose the offline distortion is $a = a(N)$ and the guard quality is constant. The two offline hierarchies in discovery give $B = O(ka^2)$. Refinement costs $O(NBa^2L) = O(Nka^4L)$, while learning center rows costs $O(N^2L/k)$. Balancing these terms gives

$$k \asymp \sqrt{N}/a^2, \qquad Q(N) = O(N^{3/2}a^2L).$$

A cubic-distortion bound $a = O(L^3)$ gives L^7, a quadratic bound $a = O(L^2)$ gives L^5, and the Räcke–Shah bound $a = O(LK)$ gives L^3K^2. The constant guard and geometric phase sum are essential to this accounting: neither contributes a hidden factor of L.

Remark 42 (The exponent is independent of the terminal sparsifier). Lemma 28 also supplies a canonical tree of height $O(\log N)$. Choose the initial hierarchy of the explicit auxiliary graph and each subsequent refinement hierarchy among such trees. Terminal compression cannot increase diameter, so the initial canonical hierarchy also has height $O(\log N)$.

The elementary guard of Remark 27 then has a uniform quality bound $\rho_0 = O(\log(2N))$. The same refinement proof, with $\gamma = (100a^2\rho_0^2)^{-1}$, costs $O(Na^2\rho_0^2B\log(2N))$ until the remaining bound satisfies $b < 1/\gamma$. The final elementary completion costs

$$O(N\rho_0^2b^2\log(2N)) = O(Na^2\rho_0^4B\log(2N)) = O(NB\log^7(2N)(\log\log(2N))^2).$$

With $B = O(k\log^2(2N)(\log\log(2N))^2)$ and $k = \lceil\sqrt{N}\rceil$, this gives $O(N^{3/2}\log^9(2N)(\log\log(2N))^4)$ queries, so the exponent $3/2$ is unchanged. The constant-quality terminal sparsifier improves the logarithmic factor.

12 Deterministic two-respecting cuts

We prove Theorem 19 using the interested-path and Monge-matrix structure of Gawrychowski, Mozes, and Weimann [GMW21]. All input-dependent measurements will be ordinary cut queries. The given tree is any tree on V, including one with edges absent from G. A cut two-respects the tree when it crosses at most two tree edges; one-edge cuts are included.

Root T at any vertex r. For a vertex v, let $T_v = \mathrm{sub}(v)$ be its set of descendants, including v. For an edge $e = (\mathrm{parent}(v), v)$, write $A_e = T_v$ and $a_e = c_G(A_e)$. Two distinct tree edges are *independent* if their subtrees are disjoint; otherwise one is a proper ancestor of the other.

Lemma 43 (Cut-query identities)**.** *For disjoint $A, B \subseteq V$,*

$$2w(A,B) = c_G(A) + c_G(B) - c_G(A \cup B). \tag{62}$$

For distinct tree edges e, f, the unique cut, up to complementation, crossing precisely e and f has the side $A_e \triangle A_f$. Its weight

$$K(e,f) = c_G(A_e \triangle A_f)$$

can therefore be evaluated with one cut query. Moreover,

$$K(e,f) = a_e + a_f - 2w(A_e, A_f) \qquad \text{if } e, f \text{ are independent,} \tag{63}$$

$$K(e,f) = a_e + a_f - 2w(A_f, V \setminus A_e) \qquad \text{if } f \text{ is below } e. \tag{64}$$

Proof. In $c_G(A) + c_G(B)$, edges between A and B are counted twice, and all other contributions are exactly $c_G(A \cup B)$, proving (62). The indicator of A_e changes along exactly the tree edge e. Taking the exclusive-or of the indicators of A_e and A_f gives the assertion about their symmetric difference. Conversely, after choosing the side containing the root, membership in a cut is determined by the tree edges it crosses, so the cut is unique up to complementation. The two displayed formulas follow by cancelling the graph edges counted twice in $a_e + a_f$. □

Query all a_e, using $n-1$ queries, and store

$$\lambda_1 = \min_{e \in E(T)} a_e \tag{65}$$

together with a minimizing edge. If $\lambda_1 = 0$, return that cut: nonnegative weights make it optimal. Henceforth assume $\lambda_1 > 0$, so every a_e is positive. We only need to ensure that every two-edge cut of weight *strictly less* than λ_1 is considered. If there is no such cut, the stored one-edge cut is already an answer. This observation handles all equality cases in the strict interest tests.

12.1 Interest paths and their deterministic discovery

12.1.1 Two boundary measures for each tree edge

For each tree edge e, put $A = A_e$ and define two nonnegative additive measures on vertex subsets:

$$\mu_e^{\mathrm{out}}(X) = w(A, X \setminus A), \tag{66}$$

$$\mu_e^{\mathrm{in}}(X) = w(A \cap X, V \setminus A). \tag{67}$$

They distribute the weight of the boundary of A over its outside endpoints and its inside endpoints, respectively. Both measures have total mass a_e. The following identities show that each evaluation costs at most two additional cut queries, since a_e is already stored:

$$2\mu_e^{\mathrm{out}}(X) = a_e + c_G(X \setminus A) - c_G(A \cup X), \tag{68}$$

$$2\mu_e^{\mathrm{in}}(X) = a_e + c_G(A \cap X) - c_G(A \setminus X). \tag{69}$$

The first follows directly from (62). For the second, apply (62) to $A \cap X$ and $V \setminus A$, and use the symmetry of the cut function.

For any nonnegative measure μ of total mass $D > 0$, call a vertex v *heavy for* μ if

$$\mu(T_v) > D/2.$$

The root is heavy. Two incomparable vertices cannot both be heavy, because their subtrees are disjoint. Also, every ancestor of a heavy vertex is heavy. Consequently the heavy vertices are precisely the vertices on a root-to-vertex path.

Let C_e be this path for μ_e^{out}, and let D_e be this path for μ_e^{in}; write c_e, d_e for their last vertices. We use the same symbols for the edge sets of these paths. Thus an edge f with lower endpoint v belongs to C_e or D_e exactly when the corresponding measure of T_v is greater than $a_e/2$. A path ending at the root has no edges.

Say that e is *interested in* f if $f \in C_e \cup D_e$. For independent edges this is exactly the cross-interest inequality from [GMW21], because

$$\mu_e^{\text{out}}(A_f) = w(A_e, A_f), \qquad \mu_e^{\text{in}}(A_f) = 0.$$

If f is below e, membership in D_e is exactly the down-interest inequality from that paper. If f is an ancestor of e, then $\mu_e^{\text{in}}(A_f) = a_e$, so $f \in D_e$ automatically. The measure definition also specifies harmless extra cross-interest tests on ancestors. Its advantage is that the two sets really are entire root paths, without any ambiguity at ancestors or at the root.

Lemma 44 (Interest captures every improvement)**.** *If $K(e, f) < \lambda_1$, then e is interested in f and f is interested in e.*

Proof. If e, f are independent, then $K(e, f) < \lambda_1 \le a_f$ and (63) imply $2w(A_e, A_f) > a_e$. Hence $f \in C_e$. Interchanging e and f gives $e \in C_f$.

If f is below e, then $K(e, f) < \lambda_1 \le a_f$ and (64) imply $2w(A_f, V \setminus A_e) > a_e$. Hence $f \in D_e$. In the other direction, $A_f \subset A_e$ gives $\mu_f^{\text{in}}(A_e) = a_f > a_f/2$, so $e \in D_f$. The other nesting order is symmetric. □

12.1.2 Finding the last heavy vertex without seeing the graph

The centroid search in [GMW21, Section 3.1] is one way of locating interest-path endpoints. In the cut-query model, the following search is convenient: it works directly on a tree of arbitrary degree and requires no auxiliary vertices or infinite-weight edges.

Lemma 45 (Majority-subtree search)**.** *Let μ be an additive nonnegative measure on V, with known total mass $D > 0$, and suppose $\mu(X)$ can be evaluated for any specified subset X. The deepest vertex c with $\mu(T_c) > D/2$ can be found deterministically with $O(\log n)$ measure evaluations.*

Proof. Fix a depth-first preorder $v_1, \dots, v_n$ of T; every rooted subtree is an interval in this order. Let $F(k) = \mu(\{v_1, \dots, v_k\})$, with $F(0) = 0$. Binary search finds the smallest k with $F(k) > D/2$ using $O(\log n)$ evaluations, since F is nondecreasing and $F(n) = D$.

Every subtree of mass greater than $D/2$ contains v_k. Indeed, if the subtree occupies the interval $[\ell, u]$, then

$$F(\ell - 1) \le D - \mu(T_v) < D/2, \qquad F(u) \ge \mu(T_v) > D/2,$$

so $\ell \le k \le u$. Its root must therefore be an ancestor of v_k. Along the known path from r to v_k, subtree masses are nonincreasing. A second binary search finds the last vertex whose subtree mass is greater than $D/2$, again in $O(\log n)$ evaluations. The root satisfies the test, so this last vertex always exists. □

Apply this lemma to each of μ_e^{out} and μ_e^{in}, using (68) and (69) for the evaluations. Thus all endpoints c_e, d_e are found with $O(n \log n)$ cut queries. Only these endpoints are stored; enumerating every edge of every interest path is unnecessary.

12.2 A small collection of path-pair problems

Construct a heavy-path decomposition $\mathcal{P}$ of the *edges* of T using only its topology. One precise convention is as follows. At each nonleaf vertex choose a child of maximum subtree size, marking the edge to that child heavy and the others light. Continue each incoming edge through the heavy outgoing edge, if one exists. The resulting maximal downward paths partition all tree edges; a nonroot path may begin with a light edge. Omit empty paths. Every root-to-vertex path intersects $O(\log n)$ paths of $\mathcal{P}$: each light edge reduces the remaining subtree size by at least a factor of two, and all consecutive heavy edges stay on one path.

For $e \in P \in \mathcal{P}$, let $\mathcal{I}(e)$ consist of the paths $Q \in \mathcal{P} \setminus \{P\}$ that contain an edge of $C_e \cup D_e$. Because C_e, D_e are root paths,

$$|\mathcal{I}(e)| = O(\log n), \qquad \sum_{e \in E(T)} |\mathcal{I}(e)| = O(n \log n). \tag{70}$$

The endpoints c_e, d_e and the decomposition determine these sets without further cut queries. They can be enumerated by jumping upward between heavy paths and deduplicating the two resulting lists.

For distinct $P, Q \in \mathcal{P}$, define the ordered lists

$$L_{P,Q} = \{e \in P : Q \in \mathcal{I}(e)\}, \qquad L_{Q,P} = \{f \in Q : P \in \mathcal{I}(f)\},$$

where each list is initially ordered from the root toward the leaves. Retain the unordered pair $\{P, Q\}$ only when both lists are nonempty. These are the *interesting path pairs*. They are constructed by producing a record (P, Q, e) for each $Q \in \mathcal{I}(e)$ and sorting/grouping the records by their path identifiers. This is computation on known data, with no oracle calls. Equation (70) gives

$$\sum_{\{P,Q\} \text{ retained}} (|L_{P,Q}| + |L_{Q,P}|) = O(n \log n). \tag{71}$$

In particular there are only $O(n \log n)$ retained pairs.

If $K(e, f) < \lambda_1$ and e, f lie on different paths P, Q, Lemma 44 ensures both that $\{P, Q\}$ is retained and that $e \in L_{P,Q}$ and $f \in L_{Q,P}$. We will search the *entire* Cartesian product of these lists. Pairs in this product need not themselves be mutually interested; including such extra pairs is harmless and allows a full Monge matrix to be searched.

12.3 Searching the pair-cut matrices

We use the convention that a matrix M is *Monge* when

$$M[i, j] + M[i', j'] \le M[i, j'] + M[i', j] \qquad (i < i',\ j < j'). \tag{72}$$

The orientations below give this sign, appropriate for finding minima.

Lemma 46 (Full Monge rectangles)**.** *Let $e_1, \dots, e_a$ and $f_1, \dots, f_b$ be two nonempty lists of tree edges, each taken from a downward path, and put $M[i, j] = K(e_i, f_j)$. The matrix is Monge in either of these cases:*

1. *All edges in one list are independent of all edges in the other, and both lists are ordered from the root toward the leaves.*

2. *Every e_i is an ancestor of every f_j, the e_i are ordered from the root toward the leaves, and the f_j are ordered toward the root.*

The lists may have gaps along their paths.

Proof. Write $A_i = A_{e_i}$ and $B_j = A_{f_j}$. In the first case, for $i < i'$ and $j < j'$ we have $A_{i'} \subset A_i$ and $B_{j'} \subset B_j$, and every A-set is disjoint from every B-set. The row and column terms in (63) cancel, leaving

$$\begin{aligned} &M[i,j] + M[i',j'] - M[i,j'] - M[i',j] \\ &\quad = -2w(A_i \setminus A_{i'}, B_j \setminus B_{j'}) \le 0. \end{aligned}$$

In the second case $A_{i'} \subset A_i$ and $B_j \subset B_{j'}$, with $B_{j'} \subset A_{i'}$. Put $R_i = V \setminus A_i$, so $R_i \subset R_{i'}$. Equation (64) reads $M[i,j] = a_{e_i} + a_{f_j} - 2w(B_j, R_i)$, and therefore

$$\begin{aligned} &M[i,j] + M[i',j'] - M[i,j'] - M[i',j] \\ &\quad = -2w(B_{j'} \setminus B_j, R_{i'} \setminus R_i) \le 0. \end{aligned}$$

Both inequalities use nonnegative graph weights. The differences are disjoint sets, and nothing requires consecutive edges in a list. □

Lemma 47 (Matrix searching: SMAWK [AKM+87])**.** *The minimum entry of a full $a \times b$ Monge matrix, together with its row and column indices, can be found deterministically using $O(a+b)$ entry evaluations.*

To apply the standard theorem, note that leftmost row-minimum indices are nondecreasing in every submatrix of a Monge matrix. Such a matrix is totally monotone for row minima. SMAWK finds those minima using $O(a+b)$ entry evaluations; comparing their values finds the minimum of the matrix. Each entry evaluation in our application is one cut query by Lemma 43.

12.3.1 Two different heavy paths

Consider a retained pair P, Q and its two lists. Split their Cartesian product into at most two full rectangles so that each rectangle is either wholly independent or wholly nested, as in Lemma 46. Here is why this is possible.

For two edge-disjoint downward paths, either all their edges are independent, or, after possibly swapping their names, an edge of P is above an edge of Q. Such an edge must be above the first edge of Q, since otherwise it would itself lie on Q. Thus a prefix of P is above every edge of Q. The remainder of P, if nonempty, is independent of every edge of Q: it has branched away from Q, since following the branch into or beyond Q would make the paths share an edge. Thus split the list from P into the ancestor part and the independent part, and pair each nonempty part with the entire list from Q. This argument also covers paths meeting at an endpoint or lying consecutively on the same branch. The split is determined by tree ancestry alone.

Orient each rectangle as in Lemma 46 and apply Lemma 47. The sum of its row and column counts, over the at most two rectangles, is at most $2(|L_{P,Q}| + |L_{Q,P}|)$. Consequently the total number of cut queries over all retained path pairs is, by (71),

$$O\left(\sum_{\{P,Q\} \text{ retained}} (|L_{P,Q}| + |L_{Q,P}|)\right) = O(n \log n). \tag{73}$$

12.3.2 Two edges on the same heavy path

Let P have k edges in root-to-leaf order. If $k \le 1$, there is nothing to search. Otherwise split P into an upper half U and a lower half W, of sizes differing by at most one. Every edge of U is an ancestor of every edge of W. Search all pairs in $U \times W$ using the nested orientation in Lemma 46 and SMAWK, at a cost of $O(k)$ cut queries. Then recurse on each half.

Every pair of distinct edges is separated at exactly one split, so it occurs in one of the searched rectangles. At each recursion depth the subpaths have total length at most k, and there are $O(\log(k+1))$ depths. This costs $O(k \log(k+1))$ queries for P, and hence

$$\sum_{P \in \mathcal{P}} O(|P| \log(|P|+1)) = O(n \log n) \tag{74}$$

over all heavy paths. This avoids searching a partial matrix or assigning artificial entries to its diagonal.

12.4 The algorithm, correctness, and query count

The complete algorithm is the following.

1. Root the given tree and compute its preorder, ancestry information, and heavy-path decomposition. Query all a_e, store a minimum one-edge cut, and return it immediately if its weight is zero.
2. Compute c_e, d_e for every edge using the two measure oracles and Lemma 45.
3. Construct the interested-edge lists and retained path pairs. Search their full Monge rectangles, retaining a minimum entry and its two defining tree edges.
4. Search pairs on each individual heavy path by the balanced recursion above, again retaining the best pair.
5. Return the lightest candidate among the stored one-edge cut and all pairs found. For a pair e, f, output $A_e \triangle A_f$; for a single edge e, output A_e.

Proof of Theorem 19. Every candidate returned by a search is a valid nontrivial cut crossing one or two tree edges, so no candidate is lighter than the optimum OPT among those cuts. If $\mathrm{OPT} = \lambda_1$, the stored one-edge cut is optimal. Otherwise an optimal cut is determined by distinct edges e, f and has $K(e,f) = \mathrm{OPT} < \lambda_1$. If e, f lie on the same heavy path, their pair occurs in a rectangle of the same-path recursion. If they lie on different paths, then Lemma 44 puts them in the two lists of a retained pair, and the splitting step puts them together in one of its rectangles. That rectangle's minimum is at most $K(e,f)$ and is itself a feasible cut. In either case the algorithm finds a cut of weight OPT.

The query bounds are

Stage	Cut queries
One-edge cut values	$n-1$
All interest-path endpoints	$O(n \log n)$
Heavy paths, lists, and rectangle splits	0
Pairs on different heavy paths	$O(n \log n)$
Pairs on the same heavy path	$O(n \log n)$
Recovering the chosen cut side	0

Their sum is $O(n \log n)$. □

12.5 All minimum cuts respecting a tree

The algorithm above discards ties on purpose: it only has to consider two-edge cuts strictly lighter than λ_1. To report every cut at a given level, three changes are needed: a search for the weakly heavy vertices of a measure, a weak form of interest that captures ties, and a listing of all minima of a Monge matrix.

12.5.1 Weak heaviness and weak interest

For a nonnegative measure μ of total mass $D > 0$, call v *weakly heavy* if $\mu(T_v) \geq D/2$. Unlike heavy vertices, two incomparable vertices can both be weakly heavy: a root with two children each carrying mass $D/2$. Every ancestor of a weakly heavy vertex is weakly heavy.

Lemma 48 (Weak majority search)**.** *Let μ be an additive nonnegative measure with known total mass $D > 0$ that can be evaluated on any specified set. The weakly heavy vertices are the vertices of two root-to-vertex paths, and their endpoints can be found deterministically with $O(\log n)$ evaluations.*

Proof. Use the preorder and the prefix masses $F(k)$ of Lemma 45. Binary search finds the least k_1 with $F(k_1) \geq D/3$ and the least k_2 with $F(k_2) \geq 2D/3$. A subtree occupying the interval $[\ell, u]$ that contains neither k_1 nor k_2 has mass less than $D/3$ if $u < k_1$, mass $F(u) - F(\ell - 1) < 2D/3 - D/3$ if $k_1 < \ell$ and $u < k_2$, and mass at most $D - F(\ell - 1) \leq D/3$ if $\ell > k_2$. Since $D > 0$, these are all below $D/2$. So the root of every weakly heavy subtree is an ancestor of v_{k_1} or of v_{k_2}. Subtree masses are nonincreasing along each root path and the root is weakly heavy, so a binary search along each of the two root paths finds the deepest weakly heavy vertex on it. □

Let C'_e and D'_e be the sets of weakly heavy vertices for μ_e^{out} and μ_e^{in}, each a union of at most two root paths, and say that e is *weakly interested in* f if the lower endpoint of f lies in $C'_e \cup D'_e$. Since this set is a union of at most four root paths, (70) holds for weak interest with a larger constant, and Lemma 48 finds the four endpoints with $O(\log n)$ cut queries per edge.

Lemma 49 (Weak interest captures every tie)**.** *If $a_e, a_f > 0$ and $K(e, f) \leq \min\{a_e, a_f\}$, then e is weakly interested in f and f is weakly interested in e.*

Proof. This is the proof of Lemma 44 with strict inequalities relaxed. If e, f are independent, $K(e, f) \leq a_f$ and (63) give $2w(A_e, A_f) \geq a_e$, that is, $\mu_e^{\text{out}}(A_f) \geq a_e/2$, so the lower endpoint of f is weakly heavy for μ_e^{out}; interchanging e and f gives the other direction. If f is below e, $K(e, f) \leq a_f$ and (64) give $2w(A_f, V \setminus A_e) \geq a_e$, that is, $\mu_e^{\text{in}}(A_f) \geq a_e/2$, while $A_f \subset A_e$ gives $\mu_f^{\text{in}}(A_e) = a_f \geq a_f/2$. □

12.5.2 All minima of a Monge matrix

Lemma 50 (Opposite minimum corners)**.** *Let M be Monge, $\ell \leq M[i, j]$ for all entries, $i < k$, $l \leq j \leq r$, and $M[i, r] = M[k, l] = \ell$. Then $M[i, j] = M[k, j]$.*

Proof. If $l = r$ then $j = l = r$ and both entries equal ℓ. Otherwise the Monge inequality for rows $i < k$ and columns $l < r$ gives $M[i, l] + M[k, r] \leq M[i, r] + M[k, l] = 2\ell$, so $M[i, l] = M[k, r] = \ell$. If $j < r$, the inequality for columns $j < r$ gives $M[i, j] \leq M[k, j]$; if $j = r$ both entries equal ℓ. Symmetrically, columns $l < j$ give $M[k, j] \leq M[i, j]$, and $j = l$ gives equality directly. □

Lemma 51 (All minima of a Monge matrix)**.** *Let M be a full $a \times b$ Monge matrix and ℓ a known number with $\ell \leq M[i, j]$ for all entries. The set of entries equal to ℓ can be listed deterministically with $O(a + b)$ entry evaluations, independently of its size.*

Proof. Find the leftmost minimum L_i of every row by SMAWK, and the rightmost minimum R_i by SMAWK applied with both the row and the column orders reversed, which keeps the Monge property (reversing only the columns would not). Evaluate $M[i, L_i]$ for every row and call the row *active* if the value is ℓ. For each column j, let A_j be the active rows with $L_i \le j \le R_i$. If $A_j \neq \varnothing$, evaluate one entry $M[i_0, j]$ with $i_0 \in A_j$ and report all of $A_j \times \{j\}$ if it equals ℓ. By Lemma 50, all entries of $A_j \times \{j\}$ are equal, so every reported entry is ℓ. Conversely, if $M[i, j] = \ell$, then row i is active and $L_i \le j \le R_i$, so (i, j) is reported. The evaluations are two SMAWK runs, a row values and at most b column representatives. □

The minima of one row need not be contiguous: two equal rows $(0, 1, 0)$ form a Monge matrix whose row minima are separated by a gap. The representative evaluation in each column is what handles this.

12.5.3 The reporting theorem

Theorem 52 (All two-respecting cuts at a lower level)**.** *Given an explicit tree T on $V(G)$ and a number $\ell > 0$ such that every cut crossing at most two edges of T has weight at least ℓ, all such cuts of weight exactly ℓ can be listed deterministically with $O(n \log n)$ cut queries, independently of their number.*

Proof. Run the algorithm of this section with three changes. First, in step 2 compute the weakly heavy paths of Lemma 48 for both measures of every edge, and build the interested-edge lists and retained path pairs from weak interest. Second, in steps 3 and 4 replace SMAWK by Lemma 51 at level ℓ in every full Monge rectangle, including those of the same-path recursion; every entry of such a rectangle is a cut crossing two tree edges, so ℓ is a lower bound on all entries, as that lemma requires. Third, report A_e for every one-edge cut whose value a_e, queried in step 1, equals ℓ, and report $A_e \triangle A_f$ for every pair (e, f) listed by Lemma 51; no candidate is queried again. Distinct listed pairs give distinct shores, since a shore crossing exactly the tree edges e and f determines the pair, so no shore is listed twice.

Every reported shore is a cut crossing at most two tree edges whose weight is exactly ℓ. Conversely, let $e \neq f$ with $K(e, f) = \ell$. Then $a_e, a_f \ge \ell = K(e, f)$ and $a_e, a_f > 0$, so Lemma 49 makes e and f weakly interested in each other. If they lie on different heavy paths, they belong to the two lists of a retained pair and the splitting step puts them in one rectangle; if they lie on the same path, the recursion separates them in one rectangle. In both cases Lemma 51 lists the pair. One-edge cuts of weight ℓ are reported directly. The query counts are those of the table above with larger constants: $O(\log n)$ evaluations per edge for the four path endpoints, and $O(a + b)$ evaluations per $a \times b$ rectangle. □

Corollary 53.

1. *With $\ell = \lambda(G) > 0$, the algorithm lists every minimum cut of G that crosses at most two edges of T.*
2. *If every cut crossing at most two edges of T has positive weight, for instance if $\lambda(G) > 0$, all cuts of least weight among them can be listed with $O(n \log n)$ queries: take ℓ to be the value found by Theorem 19.*

Proof. For the first part, every cut of G has weight at least $\lambda(G)$, so the hypothesis of Theorem 52 holds with $\ell = \lambda(G)$. For the second, the least weight ℓ of a cut crossing at most two edges is found by Theorem 19 with $O(n \log n)$ queries and is positive by assumption, so Theorem 52 applies. □

Statement on AI use

I started with the problem of finding an upper bound on the certificate complexity of global minimum cut. Before using AI, I had a proof of an $\widetilde{O}(n^{3/2})$ upper bound on certificate complexity for the complete graph, based on a two-case argument handling small and large shores separately. I gave this proof to GPT-6 Astra, which produced an $\widetilde{O}(n^{3/2})$ upper bound for a general graph, again using a two-case argument. As part of this argument, it developed an $O(n)$ certificate for the maximum s–t flow value. When I saw this, I suggested applying it within the isolating-cuts framework of Li and Panigrahi [LP25]. Astra then came up with the terminal-halving argument to obtain the $O(n \log^2 n)$ certificate.

My work on the deterministic algorithm for global minimum cut began before I used AI, in a conversation with Paweł Gawrychowski. He told me that the minimum two-respecting-cut algorithm in his joint work with Shay Mozes and Oren Weimann [GMW21] also worked deterministically in the cut query model, using $\widetilde{O}(n)$ queries. I wondered whether this could form the basis of a deterministic cut-query algorithm, rather than the flow-based approach of Anand, Saranurak, and Wang [ASW25]. I asked GPT-6 Astra to try this approach. It quickly wrote up the minimum 2-respecting cut proof in the cut query model and understood the obstacle of coming up with a small set of trees such that a minimum cut of the input graph 2-respected at least one of them. With further guided prompting, it produced the argument to construct a weak skeleton.

I used Claude Opus 5.5 and GPT-6 Astra to assist with drafting and reviewing the manuscript. The Lean formalization was done with Claude Opus 5.5 as the main driver and using GPT-6 Astra to review the code and documentation after each session.

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A Correspondence with the Lean formalization

The Lean library is at https://github.com/troyjlee/cut-query-complexity. Declaration names below are relative to the namespace `CutQueryComplexity`. Every algorithmic statement is a decision tree (`DTree`) or a query program (`Search.Comp`) selected from n and the public parameters before the hidden graph; its query cap holds for every answer function, and its correctness for every nonnegative weighted graph on the stated vertex set. Certificate statements quantify over every nonnegative weighted competitor on the same vertex set. The library's integer logarithms are $L(n) = \max\{1, \lceil \log_2(2n) \rceil\}$ (`DetMinCut.logL`) and $K(n) = L(L(n))$; the statements in natural logarithms follow from $L(n) \le 3\log(2n)$ and $K(n) \le 7\log\log(2n)$ for $n \ge 2$. Each endpoint is restated literally and audited in `CutQueryComplexityChecks/Endpoints.lean`; the only axioms in their transitive closure are `propext`, `Classical.choice` and `Quot.sound`.

The quantity $\mathrm{cert}(G)$ of (1) is `Certificate.certificateComplexity`, the least cardinality of a family of shores satisfying `Certificate.IsMinCutCertificate`; the minimum is attained (`Certificate.exists_certificate_card_eq`), and for $n \ge 2$ the empty family certifies nothing (`Certificate.one_le_certificateComplexity`). The lower bound $\mathrm{cert}(G) \le D(\lambda)$ on the deterministic cut-query complexity of the minimum-cut value is `Certificate.certificateComplexity_le_minCutValueComplexity`, with the instance-wise form `Certificate.certificateComplexity_le_cost`: a correct decision tree asks at least $\mathrm{cert}(G)$ shores on input G. Remarks 14 and 33 are not formalized: no Lean statement bounds the number of edges of a skeleton, and all constants are explicit, so no calibration is needed.

Paper	Lean	Notes

(1), $\mathrm{cert}(G)$	`Certificate.certificateComplexity`
	`Certificate.one_le_certificateComplexity`
	`Certificate.certificateComplexity_le_minCutValueComplexity`
		Least size of a value certificate; attained, $\mathrm{cert}(G) \ge 1$ for $n \ge 2$, and $\mathrm{cert}(G) \le D(\lambda)$.

Theorem 1	`Certificate.exists_minCut_witnessCertificate`
	`Certificate.exists_minCut_value_certificate`
	`Certificate.certificateComplexity_le_bound`
	`Certificate.certificateComplexity_eq_one`
		$18n\lceil \log_2(2n) \rceil^2$ shores; the witness shore is in the family. $\mathrm{cert}(G) \le 18n\lceil \log_2(2n) \rceil^2$, and $\mathrm{cert}(G) = 1$ when $\lambda(G) = 0$.

Lemma 2	`Certificate.TreePattern.public_flow_patterns`
	`Certificate.TreePattern.exists_terminal_tree_certificate`
	`Certificate.exists_cutCert`
		$4n-1$ shores fixed by a labelled tree; $4n$ for an s–t shore and value; $8n+1$ for terminal sets A, B.

Definition 3	`AllMinCuts.AllMinCutSkeleton` `TwoRespecting.IsWeakSkeleton`	The first is the definition with 111/100 in place of 5/4: $\lambda(G) \le \lambda(H)$ and $c_H(S) \le \frac{111}{100}\lambda(G)$ for every minimum cut S of G. The second is the weaker form used by the single-shore theorems, $\lambda(H) > 0$ and $4c_H(S^*) \le 5\lambda(H)$ for some minimum cut S^*, which suffices for the final search.

Paper	Lean	Notes

Theorem 4, one shore	`UndirectedClosure.deterministic_mincut_shore_value_log3loglog2`
	`DetMinCut.deterministic_mincut_log3loglog2_real_log`
	`TreeCutSparsifier.deterministic_mincut_two_stage`
		A single minimum-cut shore and the exact value; cap $Cn(\lfloor\sqrt{n}\rfloor+1)L(n)^3K(n)^2$. The last is a second proof through Theorems 6 and 7.

Theorem 4	`AllMinCuts.deterministic_all_mincuts_log3loglog2` `AllMinCuts.deterministic_all_mincuts_log3loglog2_real_log`	Output `out` with `out.value` $= \lambda(G)$ and $S \in$ `out.cuts` $\iff$ S is a minimum-cut shore (`AllMinCuts.Correct`); one decision tree per n with the same cap. When $\lambda(G) = 0$ the Lean output lists all $2^c - 2$ shores explicitly rather than the c components; the two are equivalent by `AllMinCuts.cut_eq_zero_iff_sat`. The quotient is handled by `AllMinCuts.isMinCut_iff_exists_quot` and `AllMinCuts.Correct.pullOutput`, and the final search by `AllMinCuts.coverReport_spec`; the composed program is `AllMinCuts.allTwoStageAuto_spec`.
Definition 5	`TreeCutSparsifier.TreeData.Approximates` `TreeCutSparsifier.TreeData.Good`	The second adds canonical capacities, childless terminals, and at most $8n$ nodes.
Theorem 6	`TreeCutSparsifier.exists_conversion_log3loglog2_real_log` `AllMinCuts.exists_conversion_allMinCuts`	The input is an explicit tree with arbitrary nonnegative capacities whose representation costs approximate every cut of G within B; no bound on its size is needed, and the program computes the normal form and queries the canonical capacities itself. The first returns the weaker `IsWeakSkeleton` form; the second, in integer form, returns a weak skeleton (`AllMinCuts.AllMinCutSkeleton`), with $111/100$ in place of $5/4$.

Theorem 7	`TreeCutSparsifier.theorem5LL`
	`TreeCutSparsifier.learn_tree_cut_sparsifier_LL`
	`TreeCutSparsifier.learnLL_spec`
		Integer form: $2 \cdot 3892968\, L(n)^2K(n)^2 \le \alpha$ and $\alpha^2 \le n$. The output is a safe partition and a `Good` tree of the quotient.

Lemma 8	`Discovery.residualRow_spec`	$2b\lceil\log_2(N-1)\rceil$ queries, for residual rows of $G - K$.
Lemma 9	`SupportForest.exists_support_forest` `SupportForest.exists_connectivity_test`	$3n + 4n\lceil\log_2 n\rceil \le 7nL(n)$ queries.
Remark 10	`Certificate.exists_flowCert`	$8\lvert W\rvert$ shores.
Lemma 11	`Certificate.exists_isoCert`	$9n(\lceil\log_2 \lvert R\rvert\rceil + 1)$ shores.
Lemma 12	`Certificate.exists_splitOff`	Literal.
Lemma 13	`Certificate.exists_halving` `Certificate.exists_halvingCert`	$8n$ shores.
Remark 14	—	Not formalized.

Paper	Lean	Notes
Proposition 15	`UndirectedClosure.certificate_to_skeleton` `UndirectedClosure.witnessCertificate_to_skeleton`	Zero further queries (`UndirectedClosure.sparseCompletionProg_cost`).
Lemma 16	`TwoRespecting.exists_light_spanning`	Written over ordered pairs: $(\sum_{e\in E}\ell_e)\lambda \le \sum_{a,b} w_H(a,b)\ell_{ab}$.
Lemma 17	`TwoRespecting.exists_tree_cover`	Exactly $5(\lceil\log_2 n\rceil+3)$ trees.
Remark 18	—	Commentary on Lemma 16.
Theorem 19	`TwoRespecting.twoRespecting_queryBound` `TwoRespecting.twoRespectingProg_cost`	$72n(\lceil\log_2 n\rceil+1)$ queries.
Corollary 20	`TwoRespecting.minCut_of_weakSkeleton` `TwoRespecting.weakSkeleton_minCut`	$q+5(\lceil\log_2 n\rceil+3)\big(72n(\lceil\log_2 n\rceil+1)+1\big)$ queries.
Theorem 21	`Hierarchy.Hier.supplied_weakSkeleton` with the guard `Hierarchy.Hier.ConstantCutGuard.guardFor` of `Hierarchy.constantCutGuard_42`	$\lvert\mathcal{A}\rvert\big(1+4(q-1)(\lfloor\log_2 n\rfloor+3)\big)$ queries with $q=\lceil 200\rho^2B^2\rceil$ and $\rho=42$. Promise for some minimum cut and the weaker `IsWeakSkeleton` conclusion; the pointwise form is `AllMinCuts.refine_preserves_cut`.
Lemma 22	`Discovery.quantileCheckpoints_spec`	Interval form $2M/q$ only.
Lemma 23	`Hierarchy.Hier.completion_spec`	$\eta = 2/q$; the edge count of F is not stated.
Lemma 24	`Hierarchy.Hier.cut_error_le` `Hierarchy.Hier.cut_le_rep`	Literal.
Lemma 25	`Hierarchy.treeZeroExt_42`	For hierarchies whose terminals are childless nodes.
Lemma 26	`Hierarchy.constantCutGuard_42`	$\rho_0 = 42$, literal.
Remark 27	`Hierarchy.Hier.guardFor_of_leafChildless`	Quality twice the height.
Lemma 28	`Hierarchy.RackeShah.rackeShah_hierarchyBound`	$a_0 = 268$, at most $2r+1$ nodes, height at most $2L(r)$.
Lemma 29	`Hierarchy.Hier.transfer_cut` `Hierarchy.Hier.transfer_rep`	Literal.
Lemma 30	`Hierarchy.Hier.round_promise` `Hierarchy.refine_spec`	Literal.
Theorem 31	`UndirectedClosure.supplied_hierarchy_log3loglog2` `UndirectedClosure.suppliedHierarchyProgLL_spec`	Hierarchies with at most $2n+1$ nodes and childless terminals; the program queries the canonical capacities itself. These take the promise for some minimum cut and return the weaker `IsWeakSkeleton` form; the statement of the paper, with the promise for every minimum cut, is `AllMinCuts.suppliedHierarchyProgLL_allMinCuts` in the next row. Larger supplied hierarchies are covered by `TreeCutSparsifier.exists_conversion_log3loglog2_real_log`, which has no size bound.

Paper	Lean	Notes

Remark 32	`AllMinCuts.refine_preserves_cut`
	`AllMinCuts.suppliedHierarchyProgLL_allMinCuts`
	`AllMinCuts.conversion_allMinCuts`
		Same program and budget as Theorem 31, with the pointwise argument of the remark; the constant is 111/100 in place of 5/4. `AllMinCuts.AllMinCutSkeleton.covers` is Lemma 17 applied to a weak skeleton.

Remark 33	—	Not formalized.
Lemma 34	`Discovery.tree_compression`	Literal.
Lemma 35	`Discovery.heavyNeighbor_spec` `DetMinCut.normalize_spec`	$2\lceil \log_2(N-1) \rceil$ queries per contraction; amortized total $(2\lceil \log_2 n \rceil + 1)n$.
Lemma 36	`Discovery.normalization_halves`	Literal.
Lemma 37	`Discovery.residualPrep_normalized`	$N(4 + 3\lceil \log_2 N \rceil) \cdot 4k\lceil \log_2(N-1) \rceil$ queries.
Lemma 38	`Discovery.centerSelection_normalized`	At most $\lceil (8N/k)\ln(2N) \rceil$ centers.

Lemma 39	`Discovery.psi_lower`
	`Discovery.psi_upper`
	`Discovery.auxGraph`
		Literal.

Lemma 40	`TreeCutSparsifier.certifyLL` `TreeCutSparsifier.phaseLL_spec`	The certified hierarchy is not compressed and has at most $8N$ nodes.
Remark 41, Remark 42	—	Commentary. The height-guard route is `DetMinCut.deterministic_mincut_log5`.

Lemma 43	`WeightedGraph.edge_eq_three_cuts`
	`TwoRespecting.cut_union_of_disjoint`
	`TwoRespecting.cut_sdiff_of_subset`
		Literal.

Lemma 44	`TwoRespecting.interest_indep`
	`TwoRespecting.interest_nested_below`
	`TwoRespecting.interest_nested_above`
		Literal.

Lemma 45	`TwoRespecting.majoritySearch_spec`	$4(\lfloor \log_2 n \rfloor + 1)$ queries.
Lemma 46	`TwoRespecting.pairCut_monge_indep` `TwoRespecting.pairCut_monge_nested`	Literal.
Lemma 47	`Search.smawk_spec` `Search.matrixMin_spec`	Proved rather than cited: $4\|C\| + 12\|R\|$ entry evaluations for all row minima; $4\|C\| + 13\|R\|$ for a minimum entry of the matrix.
Lemma 48	`TwoRespecting.weakMajoritySearch_spec` `TwoRespecting.weak_heavy_cover`	Stated for sums of nonnegative vertex weights; at most $8(\lfloor \log_2 n \rfloor + 1)$ evaluations.
Lemma 49	`TwoRespecting.mutual_weak_interest`	Literal.

Lemma 50, Lemma 51	`Search.monge_equal_between_opposite_minima`
	`Search.allMinima_spec`
	`Search.monge_allMinima_queryBound`
		At most $9b + 25a$ entry evaluations.

Theorem 52	`TwoRespecting.allTwoRespecting_correct_of_le` `TwoRespecting.allTwoRespecting_spec`	At most $250n(\lceil \log_2 n \rceil + 1)$ queries.
Corollary 53	`TwoRespecting.allMinTwoRespecting_spec` `TwoRespecting.allMinTwoRespecting_queryBound`	At most $322n(\lceil \log_2 n \rceil + 1)$ queries; the level comes from the existing search with no extra query.